The Journal of Chemical Physics

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Cite as: J. Chem. Phys. **150**, 024108 (2019); https://doi.org/10.1063/1.5058061
Submitted: 16 September 2018 . Accepted: 23 December 2018 . Published Online: 11 January 2019

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ABSTRACT

A position-dependent transformation is introduced for mapping a system of atomistic particles to a system of coarse-grained (CG) variables, which under some circumstances might be considered particles. This CG mapping allows atomistic particles to simultaneously contribute to more than a single CG particle and to change in time the CG particle they are associated with. That is, the CG mapping is dynamic. Mori-Zwanzig theory is then used to obtain the equations of motion for this CG mapping, resulting in conservative, dissipative, and random force terms in generalized, non-Markovian Langevin equations. In addition to the usual forces arising from the effective CG potential derived from atomistic interactions, new forces arise from the dynamic changes in the CG mapping itself. These new forces effectively account for changes arising from fluxes of atomistic particles into and out of CG ones as time progresses. Several examples are given showing the range of problems that can be addressed with this new CG mapping. These range from the usual case where atomistic particles are grouped into large molecular-like chunks, with mappings that remain fixed in time and for which an atomistic particle is part of only a single CG one, to the case where CG particles resemble fluid elements, containing many hundreds of independent atomistic particles. The new CG mapping also allows for hybrid descriptions, in which a part of the system remains atomistic or molecular-like and a part is highly coarse-grained to mesoscopic fluid element-like particles, for example. In the latter case, the equations of motion then provide the correct formalism for determining the forces, beyond the usual conservative ones. This provides a theoretical foundation upon which approximate equations of motion can be formulated to thus build numerical algorithms for expanded applications of accurate CG molecular dynamics.

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I. INTRODUCTION

Coarse-graining is the procedure by which a physical system is described by a reduced number of variables, hopefully chosen to adequately represent the quantities one is interested in. A variety of coarse-graining methods have been proposed.[1–8] Although continual increases in computing power allows ever larger systems to be simulated at an atomistic level, there are several reasons coarse-graining methods will always be needed. First, the physical behaviour of some systems, such as polymer melts, occurs on length and time scales that are large enough to be beyond fully atomistic simulation in the foreseeable future. Second, coarse-graining removes unnecessary information. For example, coarse-graining the solvent in an aqueous protein simulation not only increases computational speed but also reduces unnecessary information about atomistic solvent coordinates. The key requirement in these applications is that the coarse-graining procedure must retain the correct physical description of the system, including both structural and dynamical properties.

Many methods have been proposed to determine the coarse-grained (CG) potentials that describe interactions among CG particles.[9–13] In most cases, these interactions are

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approximated with pair potentials. In virtually all cases, a fixed CG mapping scheme is employed; that is, atoms in a complex molecule are grouped into CG particles in a chemically meaningful way, with each atom assigned to one and the same CG particle for all time. The resulting CG particles are "bonded" to each other as a result of the underlying bonding in the original molecule so that the CG procedure has a molecule-like characteristic. In other words, the atomistic molecule simply becomes a CG molecule, but for all intents and purposes is most often treated in the same manner.

There are several issues with this approach. First, the neglected degrees of freedom in the CG procedure introduce dissipative and random fluctuating terms in the equations of motion for the CG variables. A CG molecule is not simply an "atomistic" molecule composed of CG particles but fundamentally obeys different equations of motion. In principle, this could even involve non-Markovian dynamics. Formal expressions for these equations of motion can be derived using projection operator techniques.[14] One limiting case of these expressions forms the basis for dissipative particle dynamics (DPD) methods that explicitly include dissipation and thermal noise, thereby more accurately describing the CG dynamics.[15] Second, fixed CG mappings limit the scale of coarse-graining. For example, consider a simulation of a protein in an aqueous solution. Because proteins have many atoms, the usual CG procedure can be applied to produce a CG protein of bonded CG particles. What can be done with the solution though? One could coarse-grain the three atoms of water into a single CG water particle, but how does one move beyond this? If one assigns multiple water molecules to a single CG particle, the resulting object will severely distort as the underlying water molecules move independently of each other. The standard fixed CG mapping seems limited to applications only within a single molecule.

The coarse graining of unbonded particles, starting from an atomistic description, is a challenging problem. Early work on this problem dealt with deriving the equations for dissipative particle dynamics from underlying microscopic behaviour using distance based CG mappings. For example, Español *et al.*,[16] Flekkoy *et al.*,[17] and Izvekov and Rice[18] all used mappings based upon Voronoi lattices. In a similar vein, Ayton *et al.*[19] studied a mapping onto fluid blobs which incorporated a delta function to localize particles into specific blobs. More generally, Español *et al.*[20] have given a microscopic derivation of discrete hydrodynamics using discrete delta functions to map atomistic particles in a given region of space to CG variables. Using Mori-Zwanzig theory, they derived equations describing both the reversible and irreversible dynamical properties of the system. Very recently, Han *et al.*[21] studied a mapping that uses a distance-dependent weighting function to define CG particles.

The present work makes a number of advances. First, a general distance-dependent mapping is used to define CG variables. This mapping is general enough so that each CG particle can have a different mapping, if desired. This means hybrid schemes can be accommodated in which different parts of the system are coarse-grained in different ways. The mapping also allows for fractional distribution of atomistic information among the CG particles; that is, a single atom may be part of more than one CG particle. This is similar to some of the mappings of earlier studies,[17,21] except in the present case where it will be shown how such mappings require a more generalized definition for the masses of the CG particles in order to obtain a thermodynamically consistent description. Using general procedures from Mori-Zwanzig theory already discussed in the literature,[22,23] the equations of motion resulting from the general mapping are derived, including both the conservative and dissipative/random forces. These equations have many terms in them which have not been previously reported, including terms arising from changes in the mapping function and coupling between different degrees of freedom. The resulting equations provide a theoretical foundation upon which more general CG schemes can be constructed, thus expanding the range of problems that can be addressed. They also detail all the bits and pieces necessary for providing a correct description of CG particle motions.

In Sec. II, the Mori-Zwanzig theory is briefly reviewed, as it relates to the CG mapping procedure. This is applied to the standard CG mapping in Sec. III. The more general CG mapping is introduced in Sec. IV along with the corresponding equations of motion. Some examples for specific choices of the mapping are given in Sec. V followed by conclusions in Sec. VI.

II. REVIEW OF MORI-ZWANZIG THEORY

An abbreviated review of Mori-Zwanzig theory is presented below summarizing that given in Ref. 23 and the references therein. This approach derives the equations of motion for the CG variables directly. In principle, one can also seek the CG probability distribution from which equations of motion for desired variables are then derived as averages. Such an approach for the standard CG mapping is given in Ref. 24.

Consider a system of n atomistic particles whose $6n$ components of the position and momentum are represented by the trajectory $\mathbf{Z}(t)$. The time dependence of $\mathbf{Z}(t)$ is dictated by the classical Liouville equation, with the Liouville operator L being determined by the Hamiltonian, H, for the atomistic system. Let $\mathbf{z} = \mathbf{Z}(0)$ denote the initial condition of the atomistic system.

Our goal is to coarse-grain the system by defining a (hopefully small) set of M CG variables denoted $\mathbf{A}$, a vector of M components representing each CG variable. Each of these variables is a function of atomistic coordinates so that $\mathbf{A}(\mathbf{Z}(t)) = \boldsymbol{\alpha}(t, \mathbf{z})$ gives their time dependence with $\boldsymbol{\alpha} = \boldsymbol{\alpha}(0, \mathbf{z}) = \mathbf{A}(\mathbf{z})$ and where the explicit dependence of the trajectory on its initial condition has been denoted. In some contexts, these functions are called CG mappings, and the resulting variables are associated with CG particles. This nomenclature can be appropriate for CG mappings that group atoms in a molecule into larger chunks and assign the CG variables to the center of mass positions and momenta of the resulting chunks,

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for example. However, Mori-Zwanzig theory makes no such constraint, and we are free to choose general functions that produce CG variables not so easily associated with "particles." Such will be the case in this work, as these more general functions are needed if one is to move beyond the simple mapping of molecules into more globular particles, particularly so if one wants to describe collections of independent molecules.

The time dependence of $\boldsymbol{\alpha}(t, \mathbf{z})$ is given by

$$\frac{\partial \boldsymbol{\alpha}(t, \mathbf{z})}{\partial t} = L\boldsymbol{\alpha}(t, \mathbf{z}). \tag{1}$$

This equation can be rewritten so that the dependence on the CG variables are brought out explicitly, with the remaining variable dependencies dealt with in an implicit, averaged way. To accomplish this, define the conditional expectation operator, $P_{\boldsymbol{\alpha}}$, whose action on a function $F(\mathbf{z})$ is given by

$$P_{\boldsymbol{\alpha}} F = \langle F \rangle = \frac{1}{\Omega(\mathbf{z})} \int F(\mathbf{z}) \rho^{eq}(\mathbf{z}) \delta[\mathbf{A}(\mathbf{z}) - \boldsymbol{\alpha}] d\mathbf{z}, \tag{2}$$

with $\delta[x]$ being the Dirac delta function and $\rho^{eq}(\mathbf{z})$ the equilibrium probability density [in our case, we will take the canonical density, $\rho^{eq}(\mathbf{z}) = Z^{-1} \exp(-\beta H(\mathbf{z}))$ with $\beta = 1/kT$, k being the Boltzmann constant, T the temperature of the system, and Z an appropriate normalization factor], and

$$\Omega(\mathbf{z}) = \int \rho^{eq}(\mathbf{z}) \delta[\mathbf{A}(\mathbf{z}) - \boldsymbol{\alpha}] d\mathbf{z}. \tag{3}$$

Note that for a vector $\mathbf{X}$ with M components, $\delta[\mathbf{X}] = \prod_{i=1}^{M} \delta[X_i]$. Define $Q = 1 - P_{\boldsymbol{\alpha}}$, which is an operator that projects into the space orthogonal to $P_{\boldsymbol{\alpha}}$.

This operator is the one that is commonly used but is not unique. The recent work of Izvekov[25] examines this issue in some detail and proposes other choices for the projection operator. In principle, the derivations presented below could be repeated in the same manner with other projection operators. The key piece of the current proposal is the definition of the general CG mapping of Sec. IV.

Both $P_{\boldsymbol{\alpha}}$ and $\Omega(\mathbf{z})$ are equilibrium averages conditioned on $\mathbf{A}(\mathbf{z}) = \boldsymbol{\alpha}$; that is, one fixes the values of all CG variables to $\boldsymbol{\alpha}$ and then considers all atomistic states whose $\mathbf{z}$ map [that is, values of $\mathbf{A}(\mathbf{z})$] to those same fixed values. From an algorithmic point of view, one generates an equilibrium distribution of atomistic microstates (for example, with a molecular dynamics or Monte Carlo procedure), and for each, one calculates the CG variables using the functions $\mathbf{A}(\mathbf{z})$. If these calculated values match the fixed values $\boldsymbol{\alpha}$, the state is retained for calculating the integral. Otherwise, it is discarded. From another perspective, $P_{\boldsymbol{\alpha}}$ projects information from the atomistic space into the CG space. For example, if F can be written as an explicit function of CG variables only, $F = F(\mathbf{A}(\mathbf{z}))$, then $\langle F \rangle = F(\boldsymbol{\alpha})$; that is, the function is simply evaluated at the fixed values $\boldsymbol{\alpha}$. On the other hand, any F completely orthogonal to all the CG variables will give $\langle F \rangle = 0$.

Using formal operator identities and other mathematical manipulations (please consult Ref. 23 for details) Eq. (1) can be written in component form as the generalized Langevin equation

$$\frac{d}{dt}\alpha_\mu(t) = \kappa_\mu(\boldsymbol{\alpha}(t)) + \sum_\nu \int_0^t ds\, M_{\mu\nu}(\boldsymbol{\alpha}(t-s), s) \frac{\partial}{\partial \alpha_\nu} S(\boldsymbol{\alpha}(t-s)) + k \sum_\nu \int_0^t ds \frac{\partial}{\partial \alpha_\nu} M_{\mu\nu}(\boldsymbol{\alpha}(t-s), s) + R_\mu(t), \tag{4}$$

in which

$$\kappa_\mu(\boldsymbol{\alpha}) = P_{\boldsymbol{\alpha}} LA_\mu = \langle LA_\mu \rangle, \tag{5}$$

the entropy is given by

$$S(\boldsymbol{\alpha}) = k \ln \Omega(\boldsymbol{\alpha}), \tag{6}$$

the memory matrix as

$$M_{\mu\nu}(\boldsymbol{\alpha}, t) = \frac{1}{k} P_{\boldsymbol{\alpha}}(R_\mu(t) R_\nu(0)) = \frac{1}{k} \langle R_\mu(t) R_\nu(0) \rangle, \tag{7}$$

and consistent with the definition of the projection operator, the zero-mean random process (non-Gaussian in general) as

$$R_\mu(t) = \{\exp(tQL)\} QLA_\mu = Q\{\exp(tQL)\} LA_\mu. \tag{8}$$

As shown by Eq. (1),

$$LA_\mu = \frac{\partial A_\mu}{\partial \mathbf{z}} \cdot \dot{\mathbf{z}} = \dot{A}, \tag{9}$$

so κ_ν is the projection of $\dot{A}$ in the CG space. It also follows that $QLA_\mu = (1 - P_{\boldsymbol{\alpha}})LA_\mu = \dot{A} - \kappa_\mu = \delta A_\mu^Q(0)$ (with the last notation motivated by Ref. 24) that is, it gives the projection of $\dot{A}$ in the Q space. In other words, QLA_μ represents the fluctuating differences between the full $\dot{A}$ and that part of $\dot{A}$ in the CG space. Using this notation, $R_\mu(t) = \exp(tQL)\delta A_\mu^Q(0) = \delta A_\mu^Q(t)$ is the time dependence of the fluctuating differences δA_μ^Q propagated solely in the Q space. Unfortunately, it is not known in general how to generate such dynamics, and approximate expressions are usually needed. With this notation, $M_{\mu\nu} = \langle \delta A_\mu^Q(t) \delta A_\nu^Q(0) \rangle / k$ is then the projection into the CG space of the time correlation between the fluctuations (propagated in the Q space). This has the effect of a memory because it links past dynamics with the current one, and in principle, as shown by Eq. (4), this memory can extend back in time to the initial state of the system (although in practice, it usually decays to zero in a finite time). This is the fluctuation-dissipation relation for the system.

The first term on the right-hand side of Eq. (4) involves a simple projection of the dynamics in the CG space, and it is often a conservative force, the gradient of an associated effective potential (or the potential of mean force). In many CG molecular dynamics simulations, this is often the only term included in the equations of motion. The second and third terms depend upon the memory and represent the effect of past dynamics on the current one. These terms are often dissipative. The last term represents random impulses generated by the motion in the Q space on the CG variable. Overall, the

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effect of the Q space on the time dependence of the CG variable appears through the memory and random terms. Once these are known, all explicit dependence upon the Q space is removed, and Eq. (4) can then be solved (although in practice, this can be difficult owing to its stochastic nature). It is important to note the relation between the memory and the random terms in Eq. (7)—the fluctuation-dissipation relation. It is crucial that the validity of this equation is maintained in order to generate correct dynamics.

III. CG MAPPINGS

The choice for the functions $\mathbf{A}(\mathbf{z})$ is quite general but in many CG applications, it involves separate linear combinations of positions and momenta of the form

$$\begin{aligned}\mathbf{R}_K &= \sum_{i=1}^{n} c_i^K d_i^K \frac{m_i}{M_K}\mathbf{r}_i,\\ \mathbf{P}_K &= \sum_{i=1}^{n} c_i^K d_i^K \mathbf{p}_i,\end{aligned} \tag{10}$$

for a system of n atoms with $\mathbf{r}_i$, $\mathbf{p}_i$, and m_i being the position, momentum, and mass of particle i, respectively. Here $\mathbf{R}_K$ and $\mathbf{P}_K$ refer to the position and momentum variables of the CG particle K, of which there are N. The form of these equations is motivated by the work of Noid *et al.*[12] The factor d_i^K plays the role of a selection function and is non-zero only if atom i is associated with CG particle K. The coefficients c_i^K then determine the linear combinations used in forming the CG variables from the positions and momenta of the selected atoms. Note that these factors always appear as the product $c_i^K d_i^K$ in the mapping, and one could simply have assigned this to a single function. The division used here was made for conceptual reasons since it separates explicitly the selection and construction choices. Explicitly distinguishing these choices will become important when considering generalized CG mappings later.

To maintain translational invariance (that is, replacing $\mathbf{r}_i$ with $\mathbf{r}_i + \mathbf{a}$ for some constant, vector $\mathbf{a}$ should produce a corresponding shift in the CG position, $\mathbf{R}_K + \mathbf{a}$) one requires the condition

$$M_K = \sum_{i=1}^{n} c_i^K d_i^K m_i, \tag{11}$$

thereby defining the mass of CG particle K. To conserve the total momentum of the system (that is, $\sum_{K=1}^{N}\mathbf{P}_K = \sum_{i=1}^{n}\mathbf{p}_i$) one requires

$$\sum_{K=1}^{N} c_i^K d_i^K = 1 \tag{12}$$

for all i. This equation ensures that $\sum_{K=1}^{N} M_K = \sum_{i=1}^{n} m_i$ thereby conserving the total mass of the system and also guarantees that the center of mass for the entire system is the same in the atomistic and CG spaces. The usual choice is $c_i^K = 1$ in which case $\mathbf{R}_K$ and $\mathbf{P}_K$ become the position and momentum of the center of mass of CG particle K, a physically natural choice of variables.

In essentially all applications of Eqs. (10), the c_i^K and d_i^K are fixed in time, with the latter taking on binary values; that is,

$$d_i^K = \begin{cases} 1, & \text{if } i \text{ is included in CG particle } K,\\ 0, & \text{otherwise.}\end{cases} \tag{13}$$

This choice means a given atomistic particle is assigned to precisely one CG particle, and this assignment never changes in time. This corresponds to dividing atoms into groups and treating these groups as CG particles. The composition of the groups do not change in time; a situation commonly encountered when a large molecule is divided into a smaller number of CG particles (or sites).

With the choice of Eq. (13) and taking the atomistic Hamiltonian to be

$$H(\mathbf{z}) = \sum_{i=1}^{n} \frac{p_i^2}{2m_i} + U(\mathbf{r}), \tag{14}$$

with $\mathbf{r} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_n)$ representing the collection of position coordinates and U the interaction potential among the particles, Eq. (1) gives

$$L\mathbf{R}_K = \sum_{i=1}^{n} c_i^K d_i^K \frac{m_i}{M_K}\dot{\mathbf{r}}_i = \frac{\mathbf{P}_K}{M_K}, \tag{15}$$

$$L\mathbf{P}_K = \sum_{i=1}^{n} c_i^K d_i^K \dot{\mathbf{p}}_i = \sum_{i=1}^{n} c_i^K d_i^K \mathbf{f}_i = \mathbf{F}_K, \tag{16}$$

with $\mathbf{f}_i = -\partial U/\partial \mathbf{r}_i$ being the force on atom i. Since $L\mathbf{R}_K$ can be expressed solely in the CG variable space, it is unchanged under the operation of $P_{\boldsymbol{\alpha}}$ and gives zero when operated on by Q. As a consequence, and as demonstrated by Eqs. (7) and (8), all random and memory contributions are zero for any component of $\mathbf{R}_K$.

The particular choice of Eq. (13) also allows the position and momentum parts of $\Omega(\boldsymbol{\alpha})$ to be formally separated. The momentum part can be evaluated analytically (see notes in the Appendix) giving

$$\Omega(\mathbf{R},\mathbf{P}) = (2\pi kT)^{-3N/2}\left(\prod_{I=1}^{N}\widetilde{M}_I\right)^{-3/2}\exp\left[-\beta\sum_{I=1}^{N}\frac{P_I^2}{2\widetilde{M}_I} - \beta V(\mathbf{R})\right], \tag{17}$$

with $\mathbf{R} = (\mathbf{R}_1, \mathbf{R}_2, \ldots, \mathbf{R}_N)$, $\mathbf{P} = (\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N)$, $\widetilde{M}_I = \sum_{i=1}^{n}(c_i^I d_i^I)^2 m_i$, and the effective potential

$$V(\mathbf{R}) = -kT\ln\left\{\int d\mathbf{r}\,\frac{e^{-\beta U(\mathbf{r})}}{Z_r}\prod_{I=1}^{N}\delta\left[\left(\sum_{i=1}^{n} c_i^I d_i^I \frac{m_i}{M_I}\mathbf{r}_i\right) - \mathbf{R}_I\right]\right\}, \tag{18}$$

with Z_r the spatial contribution to the normalization factor Z. These results along with Eq. (6) give

$$S(\mathbf{R},\mathbf{P}) = S_0 - \frac{V(\mathbf{R})}{T} - \frac{1}{T}\sum_{I=1}^{N}\frac{P_I^2}{2\widetilde{M}_I}, \tag{19}$$

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with S_0 being a constant independent of the CG variables. With a bit more effort (see notes in the Appendix), one can also show that

$$\frac{\partial}{\partial \mathbf{R}_K} V(\mathbf{R}) = -\frac{M_K}{\widetilde{M}_K}\langle \mathbf{F}_K \rangle. \tag{20}$$

Note that the mass appearing in the canonical distribution of Eq. (17) is not the same as the mass appearing in the definition of the CG variable $\mathbf{R}_K$. This observation was also made by Noid *et al.*[12] and shows the kinds of changes that can occur in the mass definition depending upon the form of the CG variables. In the present case, both the atomistic and CG masses are constants, so their presence does not cause any formal issues, apart from being careful to distinguish them. However, in Sec. IV on dynamic CG mapping, the proper choice of mass CG variables becomes quite important.

Applying the results given above to Eq. (4) then gives the equations of motion for the CG particles as

$$\begin{aligned}
\frac{d\mathbf{R}_K}{dt} &= \frac{\mathbf{P}_K}{M_K}, \\
\frac{d\mathbf{P}_K}{dt} &= -\frac{\widetilde{M}_K}{M_K}\frac{\partial V}{\partial \mathbf{R}_K} - \beta \sum_{I=1}^{N} \int_0^t ds \langle \delta \mathbf{F}_K^Q(\mathbf{R}(t-s), s) \delta \mathbf{F}_I^Q(\mathbf{R}(t-s), 0) \rangle \\
&\quad \times \frac{\mathbf{P}_I(t-s)}{\widetilde{M}_I} + \delta \mathbf{F}_K^Q(t).
\end{aligned} \tag{21}$$

If the positions of the CG particles are taken as their centers-of-mass, then $c_i^K = 1$ and $\widetilde{M}_K = M_K$ because the use of Eq. (13) implies $(d_i^K)^2 = d_i^K$. In this case, Eqs. (21) reduce to the common ones.[23]

As Eqs. (21) shows, combining the mapping of Eqs. (10) with the choice of coefficients in Eq. (13) results in equations of motion resembling those for a particle in a bath having Newtonian conservative forces and dissipative and random fluctuating terms. The CG system behaves like a collection of particles. This meshes well with the choice of coefficients which reduce large molecules with many atoms to CG representations with a smaller number of CG particles. More importantly, the conservative forces, $\langle F_K \rangle$, determine the equilibrium structure of the CG system, while the dissipative and random forces determine the correct dynamical behaviour. If one is willing to forgo the latter, the CG equations reduce precisely to Newton's equations of motion with a new, effective CG potential, V, and so standard molecular dynamics algorithms can be used. The system is effectively treated as an "atomic" system except with CG atoms.

While the choice of coefficients in Eq. (13) is well suited for describing systems with large molecules, it has deficiencies when considering more general CG schemes. Consider a sample of liquid water. It should be possible, starting with a fully atomistic description of the system, to coarse grain to a size regime corresponding to small, macroscopic objects (fluid elements) which in turn can be described with continuum equations (Navier-Stokes equations, for example). The choice of coefficients in Eq. (13) will never give such a description, giving instead (and always) molecular-like equations of motion. The key issue is illustrated in Fig. 1 which shows in two dimensions (for simplicity) how an atomic fluid might be coarse-grained. The dashed lines divide the system into cells that resemble fluid (or volume) elements. In the left panel, atoms with the same colour have been grouped into larger CG particles using Eq. (13) at some initial time. As time progresses though, atoms move around but their assignments to the CG particles remain fixed (the colours on the atoms do not change in time). At some later time, as shown in the right panel, the CG particles no longer resemble clumps (as they did in the left panel) but instead are widely dispersed. The CG particles become stretched and expanded, and in fact given enough time, the centers of masses of all the CG particles will converge to that for the whole system. None of the atoms will actually be near the position of the CG particles. One can imagine that such a CG scheme will have great difficulty correctly describing the physics of the atomistic system.

In order to coarse grain collections of independent particles, one must move beyond the constraints of Eq. (13) and consider mappings that dynamically adjust as atoms move. For example, in the right panel of Fig. 1, one could use the grid to define the CG variables and map all particles within a given grid cell to that particle. With such a mapping, the atoms would always remain clumped within a cell, and their centers of mass would fluctuate about that for the position of the cell. Of course, atoms will also enter and exit any given cell so the number of atoms assigned to any CG particle is no longer conserved. Furthermore, the particular CG particle an atom is assigned to also changes in time. In principle, a given atom could even be assigned to multiple CG particles at the same time (say, if it was on a cell boundary, for example). This particular mapping has an Eulerian point of view since in the limit of large cells, the fluid would resemble a collection of volume elements typically used in continuum fluid dynamics. Alternatively, one could take a Lagrangian point of view and define fluid blobs relative to points that move in time, thereby producing blobs that move in space, not tethered to a particular cell. This approach has been by several authors.[17,20,21]

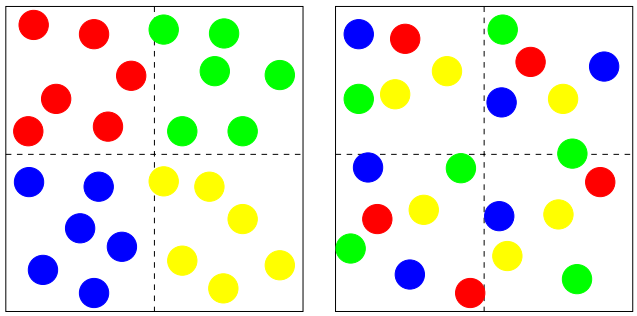


FIG. 1. Possible CG mapping scheme in which atoms are always associated with the same CG particle. The left panel shows a possible mapping of atoms into CG particles (atoms with the same colour are associated with the same CG particle) at some initial time. At some later time, the atoms have changed positions (as seen in the right panel) but their association has remained fixed, so the CG particles become distorted in time.

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One way to achieve this dynamic behaviour is to employ a mapping similar to Eq. (10) but to generalize the assignments to be functions of atomistic positions; that is, $d_i^K \equiv d_i^K(\mathbf{r})$. In this case, d_i^K becomes a continuous function of position so, for example, it could be constructed to give "all atoms within a fluid element cell of known position" the type of mapping alluded to above. This simple change greatly expands the possible CG schemes that can be employed and, in particular, allows for ones that coarse grain to the continuum limit. In Sec. IV, the equations of motion for this more generalized mapping will be derived.

IV. DYNAMIC MAPPING

A number of complications arise when using Eqs. (10) with d_i^K as a function of $\mathbf{r}$. First, Eqs. (11) and (12) must still be satisfied, so by inference c_i^K must also be a function of $\mathbf{r}$. This introduces a level of complexity we would like to avoid so we make the physically reasonable choice $c_i^K = 1$ and take the center of masses of the atoms in the CG particles. This implies, from Eq. (12), that the functions d_i^K must satisfy

$$\sum_{K=1}^{N} d_i^K = 1 \tag{22}$$

for all i. Second, M_K also becomes a function of $\mathbf{r}$, and so it is no longer constant. If one takes the usual definition $\mathbf{R}_K = M_K^{-1}\sum_{i=1}^{n} d_i^K m_i \mathbf{r}_i$ then the time dependence of $\mathbf{R}_K$ also depends upon the time dependence of M_K, which itself can be complex. While this approach is possible, we found it easier to instead use $M_K\mathbf{R}_K$ as a new CG variable. As alluded to in Sec. III, to define mappings according to cells or other points located in space, we can construct the mapping as functions of $\mathbf{r}_i - \mathbf{R}_{lab}^K$ with $\mathbf{R}_{lab}^K$ being arbitrary origins for each CG particle specified in advance and considered fixed. This also has the advantage of satisfying translational invariance since relative, rather than absolute, distances are used. Third, additional CG variables should be introduced to account for the changing masses. As shown in the Appendix and discussed below, a matrix of masses is defined that couples the distributions among pairs of particles. Fourth, since we are considering a more Eulerian point of view, the full generality of the mapping will not be considered so that d_i^K will depend only upon the position of atom i, and not upon the positions of all the other atoms. In other words, d_i^K will be a function of $\mathbf{r}_i$ rather than $\mathbf{r}$.

With these comments in mind and remembering that $d_i^K \equiv d_i^K(\mathbf{r}_i - \mathbf{R}_{lab}^K)$, the CG mapping is defined as

$$\begin{aligned}
\mathbf{W}_K &= \sum_{i=1}^{n} d_i^K m_i (\mathbf{r}_i - \mathbf{R}_{lab}^K),\\
\mathbf{P}_K &= \sum_{i=1}^{n} d_i^K \mathbf{p}_i,\\
M_{KL} &= \sum_{i=1}^{n} d_i^K d_i^L m_i,
\end{aligned} \tag{23}$$

in which, for notational simplicity, the explicit spatial dependence of d_i^K has not been explicitly shown. Note that $\mathbf{W}_K = M_K(\mathbf{R}_K - \mathbf{R}_{lab}^K)$ so $\mathbf{W}_K$ is a mass-weighted difference between the center of mass of CG particle K and its reference point, $\mathbf{R}_{lab}^K$, $\mathbf{P}_K$ is the total momentum of CG particle K, and $M_{KL} = M_{LK}$, that is, $\mathbf{M}$ is a symmetric matrix. Summing over the columns or rows of $\mathbf{M}$ gives the CG particle masses; that is, $\sum_{K=1}^{N} M_{KL} = M_L$ and $\sum_{L=1}^{N} M_{KL} = M_K$ with Eq. (11) simplifying to $M_K = \sum_{i=1}^{n} d_i^K m_i$.

The mapping of Eq. (23) is designed as a template to incorporate many different CG schemes. Each CG variable has its own function d_i^K which means different parts of the system can be coarse grained with difference schemes. For example, part of the system could be treated in the molecular way using functions like those in Eq. (13), while other parts could be coarse-grained into fluid volume elements. With the former, the values of $\mathbf{R}_{lab}^K$ could be zero so that all atoms are referenced to the same origin, and $\mathbf{W}_K$ reduces to the center of mass position of the CG particles scaled by the mass. With the latter, the values of $\mathbf{R}_{lab}^K$ could be used to reference the positions of volume elements, and $\mathbf{W}_K$ is then the displacement of the center of mass of the volume element from its reference point. In this case, changes in the density profile within the volume element are needed in order to produce a change in $\mathbf{W}_K$, so this variable is sensitive to deformations or density gradients. Equations (23) are also general enough to consider schemes where atoms contribute to more than a single CG particle at a time, and the mass matrix CG variable is specifically designed to address this case correctly. So, overall, the goal is to create a mapping with enough generality to include a range of different CG schemes, all within a common framework. The resulting equations of motion describe this entire set of possibilities. One simply needs only to specify the functions d_i^K and substitute into the resulting equations in order to obtain the correct equations of motion for that particular scheme.

One must also be mindful not to over specify the CG space. In the most general case, there are $6N + N(N+1)/2$ CG variables and $6n$ atomistic variables, and it is preferable to ensure that the former is less than the latter. Even with that condition satisfied, the linear independence of the CG variables may not be guaranteed for all choices of d_i^K since the expressions for $\mathbf{W}_K$ and M_{KL} are both functions of spatial coordinates. To guarantee linear independence, there can be no constants $a_{I,\alpha}$ and b_{IJ} satisfying

$$\begin{aligned}
&\sum_{I=1}^{N}\sum_{\alpha=x,y,z} a_{I,\alpha} W_{I,\alpha} + \sum_{I,J=1}^{N} b_{IJ} M_{IJ}\\
&\quad = \sum_{i=1}^{n}\sum_{I=1}^{N} d_i^I m_i \left[\sum_{\alpha=x,y,z} a_{I,\alpha}(r_{i,\alpha} - R_{lab,\alpha}^I) + \sum_{J=1}^{N} b_{IJ} d_i^J\right] = 0.
\end{aligned} \tag{24}$$

Applying the Liouville operator to the CG variable expressions gives

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$$
\begin{aligned}
L\mathbf{W}_K &= \sum_{i=1}^{n} d_i^K m_i \dot{\mathbf{r}}_i + \sum_{i=1}^{n}\left(\mathbf{p}_i \cdot \frac{\partial d_i^K}{\partial \mathbf{r}_i}\right)(\mathbf{r}_i - \mathbf{R}_{lab}^K) = \mathbf{P}_K + \mathbf{\Lambda}_K^1,\\
L\mathbf{P}_K &= \sum_{i=1}^{n} d_i^K\left(-\frac{\partial U}{\partial \mathbf{r}_i}\right) + \sum_{i=1}^{n}\left(\frac{\mathbf{p}_i}{m_i} \cdot \frac{\partial d_i^K}{\partial \mathbf{r}_i}\right)\mathbf{p}_i = \mathbf{F}_K + \mathbf{\Lambda}_K^2,\\
LM_{KL} &= \sum_{i=1}^{n} \mathbf{p}_i \cdot \left(d_i^L \frac{\partial d_i^K}{\partial \mathbf{r}_i} + d_i^K \frac{\partial d_i^L}{\partial \mathbf{r}_i}\right) = \mathbf{\Lambda}_{KL}^3,
\end{aligned}
\tag{25}
$$

in which we have used $\dot{\mathbf{p}}_i = -\partial U/\partial \mathbf{r}_i$ with $U = U(\mathbf{r})$ being the atomistic interaction potential. The goal is now to evaluate both the result of P_{α} acting on each of these terms and the expression for the entropy. The momentum part of these expressions can be determined analytically, leaving expressions involving the spatial parts. Details are given in the Appendix and show that

$$
\begin{aligned}
\Omega(\mathbf{W},\mathbf{P},\mathbf{M}) = (2\pi kT)^{-3N/2}|\mathbf{M}|^{-3/2}\exp\Bigg[-\frac{\beta}{2}\sum_{I,J=1}^{N}\big(&P_{I,x}M_{IJ}^{-1}P_{J,x}\\
&+ P_{I,y}M_{IJ}^{-1}P_{J,y} + P_{I,z}M_{IJ}^{-1}P_{J,z}\big) - \beta V(\mathbf{W},\mathbf{M})\Bigg],
\end{aligned}
\tag{26}
$$

in which $\mathbf{W} = (\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_N)$, $\mathbf{P} = (\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N)$, M_{IJ}^{-1} denotes the IJ-th element of the inverse of the mass matrix, and the effective potential is defined by

$$
\begin{aligned}
e^{-\beta V(\mathbf{W},\mathbf{M})} = \frac{1}{Z_r}\int d\mathbf{r}\, e^{-\beta U(\mathbf{r})}&\left\{\prod_{I=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I m_i(\mathbf{r}_i - \mathbf{R}_{lab}^I)\right) - \mathbf{W}_I\right]\right\}\\
&\times\left\{\prod_{I,J=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I d_i^J m_i\right) - M_{IJ}\right]\right\}.
\end{aligned}
\tag{27}
$$

Notice that the momentum part of $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ depends upon the mass matrix elements M_{IJ} and not upon the masses of the CG particles, M_I. It is this term that motivated us to include the mass matrix as part of the CG variable set with the definition given. Since it is part of the CG space, the entire momentum part of $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ can be evaluated analytically and be factored from the projection operator, leaving a potential term depending only upon spatial coordinates (and thereby only a function of $\mathbf{W}$ and $\mathbf{M}$). If one instead chose to include the mass of the CG particles (M_K) as the defining CG mass variables, such a factorization would not be possible and the spatial dependence of the momentum term would get included as part of the effective potential, resulting in a very complex dependence. Examining the details in the Appendix shows that the form of the momentum part of $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ arises solely from the form for the CG function for $\mathbf{P}_K$ and the fact that atoms can be part of more than a single CG particle. Thus, even if d_i^K depends upon all atomic positions $\mathbf{r}$, the form for the momentum part of $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ remains the same, implying that any general coarse graining strategies involving independent particles should use the mass matrix as part of the CG variable set.

Since $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ is the equilibrium probability distribution in the CG space, it can be used to predict averages from CG trajectories. In particular, one finds

$$
\begin{aligned}
\langle \mathbf{P}_K \cdot \mathbf{P}_L\rangle_{CG} &= 3kT\langle M_{KL}\rangle_{CG},\\
\Big\langle \sum_{K,L=1}^{N}\left[P_{Kx}M_{KL}^{-1}P_{Lx} + P_{Ky}M_{KL}^{-1}P_{Ly} + P_{Kz}M_{KL}^{-1}P_{Lz}\right]\Big\rangle_{CG} &= 3NkT,\\
\Big\langle \frac{P_K^2}{2M_K}\Big\rangle_{CG} &= \frac{3}{2}kT\Big\langle \frac{M_{KK}}{M_K}\Big\rangle_{CG},
\end{aligned}
\tag{28}
$$

in which the averages in the CG space for a function F are evaluated as

$$
\langle F\rangle_{CG} = \int F(\mathbf{W},\mathbf{P},\mathbf{M})\Omega(\mathbf{W},\mathbf{P},\mathbf{M})\, d\mathbf{W} d\mathbf{P} d\mathbf{M}. \tag{29}
$$

These expressions give the correct prescriptions for evaluating the temperature or average kinetic energy in the CG system, and show that in general, the CG momenta are correlated. This correlation is proportional to the off-diagonal elements of $\mathbf{M}$, and so they are directly related to the degree to which atoms are shared among more than a single CG particle. The greater the degree of sharing, the greater the correlation among the momenta of the CG particles involved in the sharing. This then forces the more general prescription for the kinetic energy, involving the inverse of the mass matrix and in principle involving coupling among all the CG momenta. Note in general that $\langle P_K^2/M_K\rangle_{CG}$ does not equal $3kT/2$ and $\langle \sum_{K=1}^{N} P_K^2/M_K\rangle_{CG}$ does not equal $3NkT/2$, so that usual expressions for kinetic energy do not apply when the mass matrix is non-diagonal. In fact, since $M_{KK} \leq M_K$, the value of $\langle P_K^2/M_K\rangle_{CG}$ will always underestimate $3kT/2$. Han and coworkers[21] found this explicitly for their CG mapping of fluid blobs wherein what they termed the "inertial mass" (equivalent to M_K) did not reproduce the velocity distribution at the atomistic system temperature, but a "thermal mass" did. Their definition for the "thermal mass" closely resembles the diagonal element of the mass matrix defined here. Finally, if d_i^K is constructed so that each atom i is included in only a single CG particle K, then the mass matrix is formally diagonal and $M_{KL} = M_K\delta_{KL}$. In this case, the CG momenta become uncorrelated, and the usual expressions for kinetic energy and temperature apply.

Using the expression for $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ in Eq. (6) gives the entropy as

$$
\begin{aligned}
S(\mathbf{W},\mathbf{P},\mathbf{M}) = S_0 - \frac{3}{2}k\ln|\mathbf{M}| - \frac{1}{2T}\sum_{I,J=1}^{N}\big[&P_{Ix}M_{IJ}^{-1}P_{Jx}\\
&+ P_{Iy}M_{IJ}^{-1}P_{Jy} + P_{Iz}M_{IJ}^{-1}P_{Jz}\big] - \frac{1}{T}V(\mathbf{W},\mathbf{M})
\end{aligned}
\tag{30}
$$

in which S_0 is a constant. The derivatives of the entropy with respect to the CG variables can now be explicitly evaluated as

$$
\begin{aligned}
\frac{\partial S}{\partial \mathbf{W}_K} &= -\frac{1}{T}\frac{\partial}{\partial \mathbf{W}_K}V(\mathbf{W},\mathbf{M}),\\
\frac{\partial S}{\partial \mathbf{P}_K} &= -\frac{1}{T}\sum_{I=1}^{N} M_{KI}^{-1}\mathbf{P}_I,
\end{aligned}
\tag{31}
$$

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$$\frac{\partial S}{\partial M_{KL}} = -\frac{3}{2}k(-1)^{K+L}\frac{|\widetilde{\mathbf{M}}|}{|\mathbf{M}|} - \frac{1}{T}\frac{\partial}{\partial M_{KL}}V(\mathbf{W},\mathbf{M}) - \frac{1}{2T}\sum_{I,J=1}^{N} \times \left[P_{Ix}\left(\frac{\partial M^{-1}_{IJ}}{\partial M_{KL}}\right)P_{Jx} + P_{Iy}\left(\frac{\partial M^{-1}_{IJ}}{\partial M_{KL}}\right)P_{Jy} + P_{Iz}\left(\frac{\partial M^{-1}_{IJ}}{\partial M_{KL}}\right)P_{Jz}\right],$$

in which we have used the fact that the inverse of a symmetric matrix is also symmetric, and $\widetilde{\mathbf{M}}$ is the matrix obtained by eliminating the Kth row and Lth column from $\mathbf{M}$. Note that all these equations consider the case for which $\mathbf{M}$ has only non-zero eigenvalues. If $\mathbf{M}$ has a zero eigenvalue, then the detailed expressions in the Appendix should be consulted for proper treatment, although for most physical systems this is not expected.

Expressions for the terms $P_{\boldsymbol{\alpha}}L\mathbf{W}_K$, $P_{\boldsymbol{\alpha}}L\mathbf{P}_K$, and $P_{\boldsymbol{\alpha}}LM_{KL}$ are also derived in the Appendix. Using these results in Eq. (4) gives the equations of motion for the dynamic mapping as

$$\begin{aligned}
\frac{d\mathbf{W}_K}{dt} &= \mathbf{P}_K + \sum_{I,J=1}^{N}\langle \mathbf{G}^1_{KI}\rangle \cdot M^{-1}_{IJ}\mathbf{P}_J + \frac{1}{k}\int_0^t ds\left[\sum_{I=1}^{N}\langle \delta\mathbf{W}^Q_K(s)\delta\mathbf{W}^Q_I(0)\rangle \cdot \frac{\partial S}{\partial \mathbf{W}_I} + \sum_{I=1}^{N}\langle \delta\mathbf{W}^Q_K(s)\delta\mathbf{P}^Q_I(0)\rangle \cdot \frac{\partial S}{\partial \mathbf{P}_I} + \sum_{I,J=1}^{N}\langle \delta\mathbf{W}^Q_K(s)\delta M^Q_{IJ}(0)\rangle \cdot \frac{\partial S}{\partial M_{IJ}}\right] \\
&\quad + \int_0^t ds\left[\sum_{I=1}^{N}\frac{\partial}{\partial \mathbf{W}_I}\langle \delta\mathbf{W}^Q_K(s)\delta\mathbf{W}^Q_I(0)\rangle + \sum_{I=1}^{N}\frac{\partial}{\partial \mathbf{P}_I}\langle \delta\mathbf{W}^Q_K(s)\delta\mathbf{P}^Q_I(0)\rangle + \sum_{I,J=1}^{N}\frac{\partial}{\partial M_{IJ}}\langle \delta\mathbf{W}^Q_K(s)\delta M^Q_{IJ}(0)\rangle\right] + \delta\mathbf{W}^Q_K(t), \\
\frac{d\mathbf{P}_K}{dt} &= -\sum_{I=1}^{N}M_{KI}\frac{\partial V}{\partial \mathbf{W}_I} - \sum_{I=1}^{N}\frac{\partial V}{\partial \mathbf{W}_I}\cdot\langle \mathbf{G}^1_{IK}\rangle - \sum_{I,J=1}^{N}\frac{\partial V}{\partial M_{IJ}}\langle \mathbf{G}^2_{KJI} + \mathbf{G}^2_{KIJ}\rangle + kT\sum_{I,J=1}^{N}\frac{\partial}{\partial M_{IJ}}\langle \mathbf{G}^2_{KJI} + \mathbf{G}^2_{KIJ}\rangle \\
&\quad + kT\sum_{I=1}^{N}\frac{\partial}{\partial \mathbf{W}_I}\cdot\langle \mathbf{G}^1_{IK}\rangle - kT\sum_{I,J=1}^{N}\langle \mathbf{G}^2_{IJK}M^{-1}_{IJ}\rangle + \sum_{I,I'=1}^{N}\sum_{J,J'=1}^{N}[\langle \mathbf{G}^2_{IJK}\rangle \cdot M^{-1}_{II'}\mathbf{P}_{I'}]M^{-1}_{JJ'}\mathbf{P}_{J'} \\
&\quad + \frac{1}{k}\int_0^t ds\left[\sum_{I=1}^{N}\langle \delta\mathbf{P}^Q_K(s)\delta\mathbf{W}^Q_I(0)\rangle \cdot \frac{\partial S}{\partial \mathbf{W}_I} + \sum_{I=1}^{N}\langle \delta\mathbf{P}^Q_K(s)\delta\mathbf{P}^Q_I(0)\rangle \cdot \frac{\partial S}{\partial \mathbf{P}_I} + \sum_{I,J=1}^{N}\langle \delta\mathbf{P}^Q_K(s)\delta M^Q_{IJ}(0)\rangle \cdot \frac{\partial S}{\partial M_{IJ}}\right] \\
&\quad + \int_0^t ds\left[\sum_{I=1}^{N}\frac{\partial}{\partial \mathbf{W}_I}\langle \delta\mathbf{P}^Q_K(s)\delta\mathbf{W}^Q_I(0)\rangle + \sum_{I=1}^{N}\frac{\partial}{\partial \mathbf{P}_I}\langle \delta\mathbf{P}^Q_K(s)\delta\mathbf{P}^Q_I(0)\rangle + \sum_{I,J=1}^{N}\frac{\partial}{\partial M_{IJ}}\langle \delta\mathbf{P}^Q_K(s)\delta M^Q_{IJ}(0)\rangle\right] + \delta\mathbf{P}^Q_K(t), \\
\frac{dM_{KL}}{dt} &= \sum_{I,J=1}^{N}\langle \mathbf{G}^2_{ILK} + \mathbf{G}^2_{IKL}\rangle \cdot M^{-1}_{IJ}\mathbf{P}_J + \frac{1}{k}\int_0^t ds\left[\sum_{I=1}^{N}\langle \delta M^Q_{KL}(s)\delta\mathbf{W}^Q_I(0)\rangle \cdot \frac{\partial S}{\partial \mathbf{W}_I} + \sum_{I=1}^{N}\langle \delta M^Q_{KL}(s)\delta\mathbf{P}^Q_I(0)\rangle \cdot \frac{\partial S}{\partial \mathbf{P}_I} + \sum_{I,J=1}^{N}\langle \delta M^Q_{KL}(s)\delta M^Q_{IJ}(0)\rangle \cdot \frac{\partial S}{\partial M_{IJ}}\right] \\
&\quad + \int_0^t ds\left[\sum_{I=1}^{N}\frac{\partial}{\partial \mathbf{W}_I}\langle \delta M^Q_{KL}(s)\delta\mathbf{W}^Q_I(0)\rangle + \sum_{I=1}^{N}\frac{\partial}{\partial \mathbf{P}_I}\langle \delta M^Q_{KL}(s)\delta\mathbf{P}^Q_I(0)\rangle + \sum_{I,J=1}^{N}\frac{\partial}{\partial M_{IJ}}\langle \delta M^Q_{KL}(s)\delta M^Q_{IJ}(0)\rangle\right] + \delta M^Q_{KL}(t),
\end{aligned} \tag{32}$$

in which

$$\mathbf{G}^1_{KI} = \sum_{i=1}^{n} m_i(\mathbf{r}_i - \mathbf{R}^K_{lab})d^I_i\frac{\partial d^K_i}{\partial \mathbf{r}_i}, \tag{33}$$

$$\mathbf{G}^2_{IJK} = \sum_{i=1}^{n} m_i d^I_i d^J_i\frac{\partial d^K_i}{\partial \mathbf{r}_i}, \tag{34}$$

and $\delta\mathbf{W}^Q_K = \delta\mathbf{\Lambda}^{1Q}_K$, $\delta\mathbf{P}^Q_K = \delta\mathbf{F}^Q_K + \delta\mathbf{\Lambda}^{2Q}_K$, and $\delta M^Q_{KL} = \delta\Lambda^{3Q}_{KL}$. The indices on these quantities do not indicate components. For example, for a given triple IJK, $\mathbf{G}^2_{IJK}$ is a vector with 3 components, and for a given pair KI, $\mathbf{G}^1_{KI}$ can be represented as a 3×3 matrix. Also, note that $\mathbf{G}^2_{IJK} = \mathbf{G}^2_{JIK}$. While not explicitly indicated, any CG variable dependencies in the memory or entropy terms are evaluated at the time $t - s$, as indicated by Eqs. (4).

The first thing one notices about Eqs. (32) is the complexity that results by making d^K_i a function of spatial coordinates, compared with the relative simplicity of Eqs. (21) for the case when d^K_i is constant in time. The time dependence of each CG variable has multiple dissipative terms with associated memories, along with random fluctuating terms. In addition, there are many new terms involving the quantities $\mathbf{G}^1_{KI}$ and $\mathbf{G}^2_{IJK}$. Both these quantities depend upon changes in the values of d^K_i and are zero when these coefficients are constant. So, the quantities account for the physical manifestations of atomistic particles moving from one CG particle to another. It is clear that a judicious amount of approximation and modelling will be necessary if Eqs. (32) are to become useful for computation. This process necessarily depends upon the choice of d^K_i but some general comments and observations can be made.

The terms involving memories or random forces determine the dynamics of the CG system, and the remaining terms determine the equilibrium structure. Let us consider these

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latter terms first by starting with $\mathbf{G}^1_{\mathrm{KI}}$ and $\mathbf{G}^2_{\mathrm{IJK}}$. Because Eq. (22) must always be satisfied, the change in d_i^K for any atomistic particle i must come from the values from other CG particles that i is involved with. Now, since $m_i(\mathbf{r}_i - \mathbf{R}^I_{lab})d_i^I$ is the contribution that particle i makes to the value of $\mathbf{W}_I$, $\mathbf{G}^1_{\mathrm{KI}}$ represents the contribution to the change in $\mathbf{W}_I$ as a result of particles changing their contributions to CG particle K. This is why this term (and not $\mathbf{G}^2_{\mathrm{IJK}}$) appears in the expression for $d\mathbf{W}_K/dt$. In this expression, the position of the center of mass of the CG particle changes due to its momentum, $\mathbf{P}_K$, represented by the first term, and due to a flux coming from particles changing their association with other CG particles, represented by the second term involving $\mathbf{G}^1_{\mathrm{KI}}$ multiplied by the velocity $M^{-1}_{IJ}\mathbf{P}_J$. Similarly, since $m_i d_i^I d_i^J$ is the contribution that particle i makes to the value of M_{IJ}, $\mathbf{G}^2_{\mathrm{IJK}}$ represents the contribution to the change in M_{IJ} as a result of particles changing their contributions to CG particle K. The first term on the right-hand side of the expression for dM_{KL}/dt thus represents a flux of mass contribution to M_{KL} coming from particles changing their association with other CG particles. This is the only source driving average changes in M_{KL}, as would be expected on physical grounds.

Because particles carry momentum, as well as mass, the particle fluxes mentioned above should have corresponding effects on the momentum. These are represented by the many terms involving $\mathbf{G}^1_{\mathrm{KI}}$ and $\mathbf{G}^2_{\mathrm{IJK}}$ appearing in the expression for $d\mathbf{P}_K/dt$. For example, the seventh term on the right-hand side of this expression looks to be the contribution arising from the momentum associated with the particles involved in the mass flux, while the third term is the contribution associated with the change in potential energy as a result of the mass flux. In a similar vein, the first two terms involve the usual acceleration resulting from the change in potential energy as a result of the motion of the center of mass but two different mass factors multiply it. The usual "mass of the particle" factor, represented by the first term involving M_{KI}, as well as a contribution from the particle flux, as represented by the second term involving $\mathbf{G}^1_{\mathrm{KI}}$. The fourth, fifth, and sixth terms are all multiplied by kT, implying that they are somehow related to thermal motions in the system. The physical interpretation of these terms is not clear to us at the present time.

Consider now the terms involving the memories and random forces, which satisfy the fluctuation-dissipation relation of Eq. (7). In what follows, X and Y represent any of the CG variables. The expressions in Eqs. (32) for each CG variable involve terms depending upon memories [of the form $\langle \delta \mathrm{X}^{\mathrm{Q}}(\mathrm{s}) \delta \mathrm{Y}^{\mathrm{Q}}(0)\rangle$] and terms involving derivatives of these memories. The former are multiplied by thermodynamic potentials arising from S (from partial derivatives of S), while the latter are not. Since the entropy is an extensive property, the former terms are expected to be much larger than the latter for systems of reasonable size. Thus, it is probably a reasonable approximation to ignore, in general, the terms involving derivatives of the memories, expecting their contributions to be small. The integrals involving the memory terms represent non-Markovian behaviour. However, if the values of d_i^K are well chosen, the CG variables may be "slow," meaning that their behaviour is only weakly coupled with the Q space, and a Markovian approximation can be used. In this case,

$$\int_0^t ds \langle \delta \mathrm{X}^{\mathrm{Q}}(\mathrm{s}) \delta \mathrm{Y}^{\mathrm{Q}}(0)\rangle \cdot \frac{\partial \mathrm{S}}{\partial \mathrm{Y}} \approx \gamma \cdot \frac{\partial \mathrm{S}}{\partial \mathrm{Y}}, \tag{35}$$

with

$$\gamma = \int_0^\infty ds \langle \delta \mathrm{X}^{\mathrm{Q}}(\mathrm{s}) \delta \mathrm{Y}^{\mathrm{Q}}(0)\rangle \tag{36}$$

representing a friction coefficient depending only upon the current values of the CG variables and obtained from an integral of the memory kernel. The friction coefficients result from time correlations among the fluctuations of the CG variables. In principle, correlations among the same variables, $\langle \delta \mathrm{X}^{\mathrm{Q}}(\mathrm{s}) \delta \mathrm{X}^{\mathrm{Q}}(0)\rangle$, should be significant and cannot be ignored in general. Correlations between different variables might be small or zero depending upon the choice of d_i^K. More specifically, $\delta \mathbf{W}^{\mathrm{Q}}_I$, $\delta \mathbf{P}^{\mathrm{Q}}_I$, and $\delta \mathrm{M}^{\mathrm{Q}}_{IJ}$ represent fluctuations in the mass asymmetry, momentum, and density, respectively. So, for example, $\langle \delta \mathbf{P}^{\mathrm{Q}}_I(\mathrm{s}) \delta \mathbf{P}^{\mathrm{Q}}_I(0)\rangle$ is a velocity autocorrelation proportional to the diffusion constant, and $\langle \delta \mathbf{P}^{\mathrm{Q}}_I(\mathrm{s}) \delta \mathrm{M}^{\mathrm{Q}}_{IJ}(0)\rangle$ correlates the momentum and density fluctuations.

In practice, models must be introduced to approximate the memories and other averaged quantities, such as assuming pair-wise dependencies on spatial coordinates. In principle, averages involving the projection operator can be calculated numerically using constrained molecular dynamics that generate ensembles consistent with fixed values of the CG variables. Effective potentials can be determined by a number of methods, such as iterative Boltzmann inversion or force matching.[5,27] These numerical details can only be determined once d_i^K is specified.

Finally, it is possible to obtain the rate of change of M_K from the equation for M_{KL} remembering that $M_K = \sum_L M_{KL}$. Using this relation with Eqs. (32) then gives

$$\begin{aligned}\frac{dM_K}{dt} = &\sum_{I,J=1}^{N} \langle \mathbf{G}^2_{IK}\rangle \cdot M^{-1}_{IJ}\mathbf{P}_J + \frac{1}{k}\int_0^t ds \left[\sum_{I=1}^{N} \langle \delta \mathrm{M}^{\mathrm{Q}}_K(\mathrm{s}) \delta \mathbf{W}^{\mathrm{Q}}_I(0)\rangle \cdot \frac{\partial \mathrm{S}}{\partial \mathbf{W}_I} + \sum_{I=1}^{N} \langle \delta \mathrm{M}^{\mathrm{Q}}_K(\mathrm{s}) \delta \mathbf{P}^{\mathrm{Q}}_I(0)\rangle \cdot \frac{\partial \mathrm{S}}{\partial \mathbf{P}_I} + \sum_{I,J=1}^{N} \langle \delta \mathrm{M}^{\mathrm{Q}}_K(\mathrm{s}) \delta \mathrm{M}^{\mathrm{Q}}_{IJ}(0)\rangle \cdot \frac{\partial \mathrm{S}}{\partial \mathrm{M}_{IJ}}\right] \\ &+ \int_0^t ds \left[\sum_{I=1}^{N} \frac{\partial}{\partial \mathbf{W}_I} \langle \delta \mathrm{M}^{\mathrm{Q}}_K(\mathrm{s}) \delta \mathbf{W}^{\mathrm{Q}}_I(0)\rangle + \sum_{I=1}^{N} \frac{\partial}{\partial \mathbf{P}_I} \langle \delta \mathrm{M}^{\mathrm{Q}}_K(\mathrm{s}) \delta \mathbf{P}^{\mathrm{Q}}_I(0)\rangle + \sum_{I,J=1}^{N} \frac{\partial}{\partial \mathrm{M}_{IJ}} \langle \delta \mathrm{M}^{\mathrm{Q}}_K(\mathrm{s}) \delta \mathrm{M}^{\mathrm{Q}}_{IJ}(0)\rangle\right] + \delta \mathrm{M}^{\mathrm{Q}}_K(t),\end{aligned} \tag{37}$$

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in which Eq. (22) has been used to give $\sum_L \langle G^2_{IKL} \rangle = 0$ and

$$\mathbf{G}^2_{IK} = \sum_{i=1}^{n} m_i d_i^I \frac{\partial d_i^K}{\partial \mathbf{r}_i}. \tag{38}$$

These equations show that indeed some of the terms reduce to dependencies upon M_K alone but others still retain an explicit dependence upon M_{KL}. This shows why it is necessary to use the full mass matrix in general. However, if the terms depending upon M_{KL} are modelled or approximated in some way, Eq. (37) might be useful in some circumstances.

V. CHOICES FOR d_i^K

Equations (32), in principle, prescribe the equations of motion for predicting the correct equilibrium structure and dynamics of the CG system for any choice of $d_i^K(\mathbf{r}_i)$. In this section, several specific choices will be presented and discussed to give a flavour of the types of CG schemes that are possible within the general framework.

A. Constant d_i^K

With this case, the values of d_i^K are chosen to be fixed, constant in time. This is the CG mapping, as shown in the left panel of Fig. 2, typically applied to molecules. Thus, any spatial derivatives of d_i^K give zero so that $\mathbf{\Lambda}^1_K = \mathbf{\Lambda}^2_K = \mathbf{\Lambda}^3_{KL} = 0$ from which it follows that $\delta \mathbf{W}^Q_K = \delta M^Q_{KL} = 0$ and $\delta \mathbf{P}^Q_K = \delta \mathbf{F}^Q_K$. This has the effect of removing all random fluctuating terms except those for $\mathbf{P}_K$, as well as any memory terms involving either $\mathbf{W}_K$ or M_{KL}. In addition, this case gives $\mathbf{G}^1_{KI} = \mathbf{G}^2_{IJK} = 0$ which eliminates many terms from the general equation and specifically gives $dM_{KL}/dt = 0$; that is, the mass matrix is constant. This is expected since fixed d_i^K implies fixed mass distribution among the CG particles. Employing all these simplifications, along with $\mathbf{R}^K_{lab} = 0$ reduces Eqs. (32) to

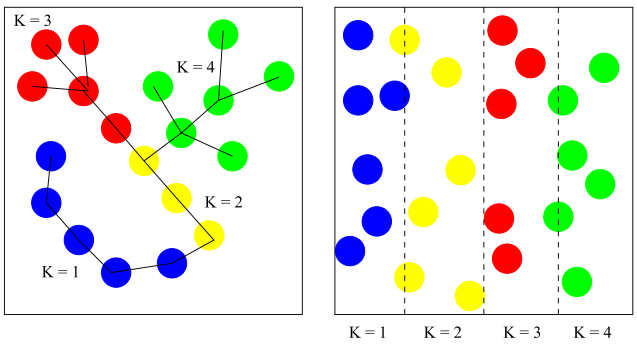


FIG. 2. The left panel shows a standard CG mapping by grouping atoms within a molecule into larger CG particles. The right panel shows the system divided into slabs with each slab identified with a CG particle. Atoms falling within each slab are assigned to the corresponding CG particle, and as atoms move, their assignments are changed dynamically according to the slab they occupy. Atoms with the same colour are associated with the same CG particle which in turn are labelled $K = 1$–4.

$$\begin{aligned}
\frac{d\mathbf{W}_K}{dt} &= \mathbf{P}_K, \\
\frac{d\mathbf{P}_K}{dt} &= -\sum_{I=1}^{N} M_{KI} \frac{\partial V}{\partial \mathbf{W}_I} - \beta \sum_{I,J=1}^{N} \int_0^t ds \\
&\quad \times \langle \delta \mathbf{F}^Q_K(s) \delta \mathbf{F}^Q_I(0) \rangle \cdot M^{-1}_{IJ} \mathbf{P}_J + \delta \mathbf{F}^Q_K(t), \\
\frac{dM_{KL}}{dt} &= 0.
\end{aligned} \tag{39}$$

The factors of M_{KL} (although constant) account for the possibility of non-integral values of d_i^K, indicating that contributions from an atom are distributed among more than a single CG particle. In such a situation, $\mathbf{M}$ has non-zero off-diagonal elements. However, if the common CG mapping scheme of Eq. (13) is used then $d_i^I d_i^J = d_i^I \delta_{IJ}$ since, for a given atom i, there is only a single CG particle giving a non-zero value of d_i^I. As well, this non-zero value must be one so that $(d_i^I)^n = d_i^I$ for any integer multiple n. With this simplification, $M_{IJ} = M_I \delta_{IJ}$ and Eqs. (39) reduce to Eqs. (21) once one recalls that $\mathbf{W}_K = M_K \mathbf{R}_K$. Thus, the general equations reduce to the correct ones in this limit.

B. Binary d_i^K

In this case, it is imagined that every atom is assigned to only a single CG particle; however, this assignment may change in time. Thus, d_i^K has a value of 0 or 1, but the particular value depends upon the position of the atom. Essentially, the value instantaneously changes from one value to another. Because an atom can only be associated with one CG particle at a time, $d_i^I d_i^J = d_i^I \delta_{IJ}$ (since $(d_i^I)^2 = d_i^I$ in this case) from which follows $M_{IJ} = M_I \delta_{IJ}$, $M^{-1}_{IJ} = M^{-1}_I \delta_{IJ}$, $|\mathbf{M}| = \prod_{I=1}^N M_I$, $\mathbf{G}^1_{KI} = \mathbf{G}^1_K \delta_{IK}$, and $\mathbf{G}^2_{IJK} = \mathbf{G}^2_K \delta_{IJ} \delta_{JK}$, with

$$\mathbf{G}^1_K = \sum_{i=1}^{n} m_i (\mathbf{r}_i - \mathbf{R}^K_{lab}) d_i^K \frac{\partial d_i^K}{\partial \mathbf{r}_i}, \tag{40}$$

$$\mathbf{G}^2_K = \sum_{i=1}^{n} m_i d_i^K \frac{\partial d_i^K}{\partial \mathbf{r}_i}. \tag{41}$$

The expression for the entropy simplifies to

$$S(\mathbf{W}, \mathbf{P}, \mathbf{M}) = S_0 - \frac{3}{2} k \sum_{I=1}^{N} \ln M_I - \frac{1}{2T} \sum_{I=1}^{N} \frac{P_I^2}{M_I} - \frac{1}{T} V(\mathbf{W}, \mathbf{M}). \tag{42}$$

Considering the above results leads to several simplifications in the general Eq. (32). First, terms involving summations of mass matrix elements simplify to either single terms or in some cases summations over M_I. For example, the non-dissipative and non-random terms for $\mathbf{W}_K$ become

$$\langle L \mathbf{W}_K \rangle = \mathbf{P}_K + \langle \mathbf{G}^1_K \rangle \cdot \frac{\mathbf{P}_K}{M_K}, \tag{43}$$

with the second term on the right-hand side accounting for the flux of contributions to $\mathbf{W}_K$ resulting from atoms changing their contributions to CG particles as time progresses.

Similar simplifications occur with the expressions for $d\mathbf{P}_K/dt$ and dM_K/dt. Unfortunately, the memory and random fluctuation terms in the general Eq. (32) mostly carry over in

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this case, so the simplification of these parts of the general equations is limited.

The right panel of Fig. 2 shows a specific example using this case where the atomistic system is divided into slabs of width ℓ along the x-direction. An atom is assigned to the slab it occupies, so as it moves in space, this assignment changes. On the boundary between slabs, assign $d_i^K = 1$ if the particle is on the left boundary of the slab and $d_i^K = 0$ if on the right. In such a CG scheme, $d_i^K = H(x_i - (K-1)\ell, 1)H(K\ell - x_i, 0)$ with the step function defined as

$$H(x,a) = \begin{cases} 1, & x > 0, \\ a, & x = 0, \\ 0, & x < 0, \end{cases} \tag{44}$$

and K labelling a slab. We expect that the relative distance from a slab will be the important parameter rather than an absolute distance, and so in this case, it seems reasonable to set $\mathbf{R}_{lab}^K = (K-1)\ell\hat{x}$, that is, the left most boundary of slab K in the x-direction. This then gives

$$\begin{aligned} d_i^K \frac{\partial d_i^K}{\partial \mathbf{r}_i} &= H(x_i - R_{lab,x}^K, 1)H(R_{lab,x}^K + \ell - x_i, 0) \\ &\quad \times \left\{ \delta[x_i - R_{lab,x}^K] H(R_{lab,x}^K + \ell - x_i, 0) \right. \\ &\quad \left. - \delta[R_{lab,x}^K + \ell - x_i] H(x_i - R_{lab,x}^K, 1) \right\} \\ &= \delta[x_i - R_{lab,x}^K] \end{aligned} \tag{45}$$

so that $\mathbf{G}_K^1$ and $\mathbf{G}_K^2$ are zero except when an atom crosses the left boundary of a slab. This effectively imposes another constraint on the conditional expectation value by selecting only those configurations which in addition to having fixed CG values, also have an atom with an x-component value at the left boundary of slab K.

C. Lattice d_i^K

The idea of the previous case can be extended by considering a CG scheme that divides a system into a lattice, as shown by the dashed lines in Fig. 1. Consider a cubic lattice with edge length 2ℓ in which $\mathbf{R}_{lab}^K$ points to the center of lattice K. To construct a CG scheme resembling volume elements, one would assign all atoms within cube K to CG particle K. Here the boundary issues become more complex because two, four, or eight cubes can meet at a boundary. A symmetric construction of d_i^K would be

$$d_i^K = \begin{cases} 1, & \text{if atom } i \text{ is within cube } K, \\ 1/2, & \text{if atom } i \text{ is on a face of cube } K, \\ 1/4, & \text{if atom } i \text{ is on an edge of cube } K, \\ 1/8, & \text{if atom } i \text{ is on a vertex of cube } K, \\ 0, & \text{otherwise,} \end{cases} \tag{46}$$

and a function accomplishing this would be

$$\begin{aligned} d_i^K &= H(x_i - R_{lab,x}^K + \ell, 1/2)H[R_{lab,x}^K + \ell - x_i, 1/2) \\ &\quad \times H(y_i - R_{lab,y}^K + \ell, 1/2)H(R_{lab,y}^K + \ell - y_i, 1/2) \\ &\quad \times H(z_i - R_{lab,z}^K + \ell, 1/2)H(R_{lab,z}^K + \ell - z_i, 1/2). \end{aligned} \tag{47}$$

This case looks very similar to the previous case, except $\mathbf{M}$ is not strictly diagonal because particles on a cube boundary are assigned to multiple CG particles at the same time. This means off-diagonal elements of $\mathbf{M}$ will be non-zero. This also implies there will be non-zero elements of $\mathbf{G}_{KI}^1$ when $K \neq I$ and of $\mathbf{G}_{IJK}^2$ for $I \neq J \neq K$ because atoms on an edge or vertex are assigned to more than 3 CG particles. Similar to the previous case though, derivatives of d_i^K will produce delta functions located at the cube boundaries which have the effect of further constraining the conditional expectation values.

The selection function of Eq. (47) creates a CG scheme resembling volume elements, so in principle it could make contact with continuum theory if the cubes contained enough atoms. The consequence of this choice is that the physics of the system is necessarily condensed to boundary terms through the $\mathbf{G}_{KI}^1$ and $\mathbf{G}_{IJK}^2$ expressions and thus to nearest neighbours. An alternative is to consider a selection function that extends to much longer ranges, for example,

$$d_i^K = \frac{1}{N_K(\mathbf{r})} e^{-a|\mathbf{r}_i - \mathbf{R}_{lab}^K|}, \tag{48}$$

with $N_K(\mathbf{r}) = \sum_{I=1}^n \exp(-a|\mathbf{r}_i - \mathbf{R}_{lab}^I|)$ chosen to satisfy Eq. (22) and a a constant that determines an effective range. This function effectively includes all atoms in every CG particle but with an exponential weighting. Thus, those atoms closest to $\mathbf{R}_{lab}^K$ (the "center" of CG particle K) are weighted the most. Such a function allows for the description of long-range interactions where particles distant from a CG one can still exert some influence. This function also has a simple derivative, $\partial d_i^K / \partial \mathbf{r}_i = a d_i^K (d_i^K - 1)\widehat{(\mathbf{r}_i - \mathbf{R}_{lab}^I)}$ with the hat denoting a unit vector, leading to simple expressions for $\mathbf{G}_{KI}^1$ and $\mathbf{G}_{IJK}^2$. Unfortunately, because of its more general nature, no analytic simplifications result in Eq. (32) using this function. As well, the normalization factor $N_K(\mathbf{r})$ constantly changes thus necessitating even more complex algorithms.

The sharp boundaries of Eq. (47) look appealing since they produce CG particles with well-defined volumes and close conceptual connections with volume elements used in computational fluid dynamics. However, work by Eriksson *et al.* has shown that such sharp boundaries lead to discontinuous changes in particle momenta, which causes numerical issues in the calculation of CG properties.[26] Thus, from a numerical point of view, it is likely better to use functions that grow quickly, yet continuously, across the cell boundary (like the arctangent function). This produces a finite width to the crossing region which then eliminates discontinuous changes in the CG variables but then necessarily introduces non-zero off-diagonal elements in the mass matrix. If the crossing region is not too large, the off-diagonal elements may be small enough to ignore or approximate in some way.

D. Hybrid schemes

The cases considered up to this point have all treated the system in a homogeneous way, applying the same CG scheme

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to all particles. However, this is not required as Eqs. (32) allow for quite flexible schemes, including those where parts of the system are treated differently. For example, imagine that a part of the system is treated as in the "Constant d_i^K" case and a part as in the "Lattice d_i^K" case. We might want to do this when describing a protein in solution, for example, with the "Constant d_i^K" scheme applied to the atoms in the protein and the "Lattice d_i^K" scheme applied to the solution.

More specifically, let $K = 1, \ldots, N_1$ be "Constant d_i^K" CG particles involving atoms $i = 1, \ldots, n_1$, and let $K = N_1 + 1, \ldots, N$ be "Lattice d_i^K" CG particles involving atoms $i = n_1 + 1, \ldots, n$. Building on the results of the previous cases, in the hybrid scheme $\mathbf{M}$ will divide into two blocks, one diagonal with elements M_K for $K = 1, \ldots, N_1$ and the other non-diagonal. The matrix $\mathbf{M}^{-1}$ is divided similarly with the diagonal part having elements $1/M_K$. Since $\mathbf{\Lambda}^1_K = \mathbf{\Lambda}^2_K = \mathbf{\Lambda}^3_{KL} = 0$ for $K, L = 1, \ldots, N_1$, the equations of motion for the "Constant d_i^K" CG particles will contain no memory or random fluctuation terms for the $\mathbf{W}_K$ and M_{KL} variables and reduce to those with $\mathbf{F}_K$ for the $\mathbf{P}_K$ variables. Furthermore, $\mathbf{G}^1_{KI} = \mathbf{G}^2_{IJK} = 0$ if any of I, J, or K is in the range $1, \ldots, N_1$. Using all this information with Eqs. (32) then gives the equations of motion for $K = 1, \ldots, N_1$ as

$$
\begin{aligned}
\frac{d\mathbf{W}_K}{dt} &= \mathbf{P}_K,\\
\frac{d\mathbf{P}_K}{dt} &= -M_K\frac{\partial V}{\partial \mathbf{W}_K} - \beta\sum_{I=1}^{N_1}\int_0^t ds\,\langle \delta\mathbf{F}^Q_K(s)\delta\mathbf{F}^Q_I(0)\rangle\cdot\frac{\mathbf{P}_I}{M_I} + \delta\mathbf{F}^Q_K(t)\\
&\quad + \frac{1}{k}\int_0^t ds\left[\sum_{I=N_1+1}^{N}\langle \delta\mathbf{F}^Q_K(s)\delta\mathbf{W}^Q_I(0)\rangle\cdot\frac{\partial S}{\partial \mathbf{W}_I} + \sum_{I=N_1+1}^{N}\langle \delta\mathbf{F}^Q_K(s)\delta\mathbf{P}^Q_I(0)\rangle\cdot\frac{\partial S}{\partial \mathbf{P}_I} + \sum_{I,J=N_1+1}^{N}\langle \delta\mathbf{F}^Q_K(s)\delta M^Q_{IJ}(0)\rangle\cdot\frac{\partial S}{\partial M_{IJ}}\right]\\
&\quad + \int_0^t ds\left[\sum_{I=N_1+1}^{N}\frac{\partial}{\partial \mathbf{W}_I}\langle \delta\mathbf{F}^Q_K(s)\delta\mathbf{W}^Q_I(0)\rangle + \sum_{I=N_1+1}^{N}\frac{\partial}{\partial \mathbf{P}_I}\langle \delta\mathbf{F}^Q_K(s)\delta\mathbf{P}^Q_I(0)\rangle + \sum_{I,J=N_1+1}^{N}\frac{\partial}{\partial M_{IJ}}\langle \delta\mathbf{F}^Q_K(s)\delta M^Q_{IJ}(0)\rangle\right],\\
\frac{dM_{KL}}{dt} &= 0.
\end{aligned}
\tag{49}
$$

For $K = N_1 + 1, \ldots, N$, the equations of motion are essentially the same as those in Eqs. (32), except with many of the summations running only over the "Lattice d_i^K" CG particles. Comparing Eqs. (39) and (49) shows that the effect of the "Lattice d_i^K" CG particles enters in two ways: (i) implicitly through the variable dependence in the effective potential V, and (ii) explicitly through memory and random fluctuations terms specifically linking the two different particle descriptions. Presumably, these effects become more significant when the two different types of particles approach one another. The important point though is that the formulation gives a prescription for the correct description of the system including both conservative and dynamical terms. No *ad hoc* procedures are necessary for dealing with the interaction between the two different CG types. In principle, many more complex hybrid CG schemes could be devised for which Eqs. (32) will ultimately yield the correct equations of motion.

VI. CONCLUSIONS

The goal of the present work was to propose a framework in which more general CG schemes could be formulated. The idea put forward is quite simple, namely, allowing the selection function in the CG mapping, d_i^K, to be a function of an atom's position. This allows for CG schemes that divide a system into spatial regions, like volume elements used in continuum fluid simulations, for example. More generally, an atom can be fractionally assigned to CG particles, and this assignment is allowed to change in time, depending upon the position of the atom. The framework treats the case in which only the position of a given atom determines its CG assignment, that is, $d_i^K \equiv d_i^K(\mathbf{r}_i)$ and $\mathbf{R}^K_{lab}$ are fixed origins. This is adequate for a Eulerian description of the system if a cell-based CG scheme is employed. This can be generalized by making the assignment for particle i a function of all atomistic positions, that is, $d_i^K \equiv d_i^K(\mathbf{r})$. In this case, internal reference points could be used as origins for CG particle assignments, and a Lagrangian description could be formulated. Such a generalization would lead to equations very similar to Eqs. (32) but with slightly more complicated definitions for $\mathbf{G}^1_{KI}$ and $\mathbf{G}^2_{IJK}$. Note that Eqs. (28) remain the same even with this generalization. More generally though, a mixed Lagrangian and Eulerian description is possible in a hybrid scheme with molecular-like functions for d_i^K being used for part of the system (Lagrangian) and cell-based ones for another (Eulerian).

A number of general comments and observations can be made from the equations of motion for the framework, Eqs. (32). First, the equations are quite complex, involving memory and random fluctuation terms arising from all possible correlations in the system. The effective potential is also generalized. In addition, two new quantities arise from the theory, $\mathbf{G}^1_{KI}$ and $\mathbf{G}^2_{IJK}$, that account for the physical consequences of the changes in d_i^K, that is, for atoms

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moving between different CG particles. These quantities also link changes in pairs and triples of CG particles. In short, the coarse-graining of *independent* particles introduces complexity not only in the mathematical description of the motion but also in the physical space since the movement of atoms between CG particles causes fluxes of mass and momentum, which in turn produce a range of physical effects that need to be correctly accounted for.

Second, CG schemes that allow atoms to be shared among more than a single CG particle introduce, as evidenced by Eqs. (28), another layer of complexity. In this case, the shared atoms cause the mass and momenta of the corresponding CG particles to become correlated. This means the mass of a CG particle is no longer an independent quantity thus forcing the use of the more general mass matrix. Since CG particle momenta are also not independent, the calculation of kinetic energies and the system temperature requires the more general expressions given by Eqs. (28). While it is possible to choose d_i^K to map atoms to single CG particles only, the discontinuous changes in momentum and mass resulting from such choices seem to cause numerical difficulties. So, in practice, it appears necessary to smooth the boundaries between CG particles, meaning some degree of atom sharing is necessary, and thus the mass matrix is unavoidable. It is clear that more work is necessary to explore ways in which the mass matrix can be dealt with efficiently since its use not only complicates the equations of motion but also introduces many more CG variables whose rate of change needs to be determined, in principle.

Third, as atoms move among CG particles, $\mathbf{G}^1_{\mathrm{KI}}$ and $\mathbf{G}^2_{\mathrm{IJK}}$ account for the resulting flux of momentum, mass and density variations. These terms are formally independent of the effective potential and instead depend upon the choice of d_i^K. So, it should be possible to model them in a system-independent manner. The behaviour of these quantities and their modelling will be a focus of future work.

Fourth, while Eqs. (32) provide the theoretical foundation for the correct equations of motion, describing both equilibrium structure and dynamics, in practice approximations will be needed to make them tractable. As is usually done, the effective potential could be assumed to be pair-wise additive. If the CG variables change on a slow enough time scale, then a Markovian approximation can be used to replace the integrals over the memory functions with friction-like terms. In this limit, Eqs. (32) would transform into dissipative particle dynamics (DPD)-like equations. The same techniques used to model effective CG potentials and memories can be applied in the present case, except with a bit more complex behaviour. Ultimately, one is approximating many-body interactions with simpler models designed, in the best case scenario, to capture the essential behaviour of the system.

In addition to these general comments, a number of examples were elaborated for different choices of d_i^K, ranging from fixed values typically used in many CG schemes to date, to more complex hybrid schemes that simultaneously employ several different d_i^K assigning strategies. These examples demonstrate the flexibility of the new framework for designing more complex CG schemes and allow for a description of many physical and chemical systems. Essentially, one needs only to specify the functions d_i^K, presumably chosen to best describe a problem of interest, and then substitute these functions into the general equations to obtain the equations of motion for that particular CG scheme. Once d_i^K is specified, the definitions for all the remaining quantities become explicit, and the task of obtaining the effective potential and memory or friction terms can be done.

Overall, Eqs. (32) provide a theoretical foundation, which with the judicious use of approximations, allows for the correct description of a general class of CG mapping schemes. Future work is needed to elaborate many of the details associated with this process. However, in time, this should lead to broader applications of CG simulation methodologies, especially those employing hybrid CG schemes.

ACKNOWLEDGMENTS

The authors would like to thank Pep Español for helpful and stimulating discussions. This study was supported by a grant from the Natural Sciences and Engineering Research Council (NSERC) of Canada.

APPENDIX: DERIVATION OF EQUATIONS OF MOTION

We begin the derivation of Eqs. (32) by explicitly evaluating the momentum contributions in $P_{\boldsymbol{\alpha}}$. Considering the general expression of Eq. (2) using the mapping of Eqs. (23) with a Hamiltonian of Eq. (14) define

$$\begin{aligned}\mathcal{P}\mathrm{F}(\mathbf{r},\mathbf{p}) = \frac{1}{Z}\int d\mathbf{r}\, e^{-\beta \mathrm{U}(\mathbf{r})}&\left\{\prod_{I=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d_i^I m_i(\mathbf{r}_i - \mathbf{R}^I_{lab})\right) - \mathbf{W}_I\right]\right\}\\ &\times\left\{\prod_{I,J=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d_i^I d_i^J m_i\right) - \mathrm{M}_{IJ}\right]\right\}\\ &\times\int d\mathbf{p}\, \mathrm{F}(\mathbf{r},\mathbf{p}) e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{I=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d_i^I \mathbf{p}_i\right) - \mathbf{P}_I\right]\right\},\end{aligned} \tag{A1}$$

in which all the terms depending upon $\mathbf{p}$ have been isolated in the last term, and $Z = (2\pi kT)^{3n/2}(\prod_{i=1}^{n} m_i)^{3/2} Z_r$. With this definition, $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ is $\mathcal{P}$ operating on $\mathrm{F} = 1$, and $P_{\boldsymbol{\alpha}}\mathrm{F} = \langle \mathrm{F}\rangle = \mathcal{P}\mathrm{F}/\Omega$. It will prove useful to first explicitly evaluate the following integral:

$$I^{\ell}_{j,x} = \int_{-\infty}^{\infty} d\mathbf{p}_x\, p^{\ell}_{j,x} e^{-\beta\sum_{i=1}^{n}\frac{p_{i,x}^2}{2m_i}}\left\{\prod_{I=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d_i^I p_{i,x}\right) - \mathrm{P}_{I,x}\right]\right\}, \tag{A2}$$

for $\ell = 0, 1, 2$. Rewriting the delta function using $\delta[x] = (2\pi)^{-1}\int_{-\infty}^{\infty} d\omega\ \exp[i\omega x]$ and collecting like terms then gives

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$$
\begin{aligned}
I^{\ell}_{j,x} &= \int_{-\infty}^{\infty} d\mathbf{p}_x\, p^{\ell}_{j,x} e^{-\beta \sum_{i=1}^{n} \frac{p^2_{i,x}}{2m_i}} \left\{ \prod_{I=1}^{N} \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega_I \exp\left(i\omega_I \left[\left(\sum_{i=1}^{n} d^I_i p_{i,x} \right) - P_{I,x} \right] \right) \right\} \\
&= \left(\frac{1}{2\pi} \right)^N \int_{-\infty}^{\infty} d\boldsymbol{\omega}\; e^{-i \sum_{I=1}^{N} \omega_I P_{I,x}} \int_{-\infty}^{\infty} d\mathbf{p}_x\; e^{-\beta \sum_{i=1}^{n} \frac{p^2_{i,x}}{2m_i}} p^{\ell}_{j,x} e^{i \sum_{i=1}^{n} \left(\sum_{I=1}^{N} \omega_I d^I_i \right) p_{i,x}} \\
&= \left(\frac{1}{2\pi} \right)^N \int_{-\infty}^{\infty} d\boldsymbol{\omega}\; e^{-i \sum_{I=1}^{N} \omega_I P_{I,x}} \left\{ \prod_{i \neq j} \int_{-\infty}^{\infty} dp_{i,x}\; e^{-\beta \frac{p^2_{i,x}}{2m_i} + i \left(\sum_{I=1}^{N} \omega_I d^I_i \right) p_{i,x}} \right\} \int_{-\infty}^{\infty} dp_{j,x}\; p^{\ell}_{j,x} e^{-\beta \frac{p^2_{j,x}}{2m_j} + i \left(\sum_{i=1}^{n} \omega_I d^I_j \right) p_{j,x}},
\end{aligned} \tag{A3}
$$

in which $d\boldsymbol{\omega} = \prod_{I=1}^{N} d\omega_I$. The integrals over momentum can be evaluated analytically by completing the square in the exponents, giving

$$
\begin{aligned}
I^{\ell}_{j,x} &= \left(\frac{1}{2\pi} \right)^N \int_{-\infty}^{\infty} d\boldsymbol{\omega}\; e^{-i \sum_{I=1}^{N} \omega_I P_{I,x}} \left\{ \prod_{i \neq j} \sqrt{\frac{2\pi m_i}{\beta}} e^{-\frac{m_i}{2\beta} \left(\sum_{I=1}^{N} \omega_I d^I_i \right)^2} \right\} \sqrt{\frac{2\pi m_j}{\beta}} F^{\ell}_j(\boldsymbol{\omega}) e^{-\frac{m_j}{2\beta} \left(\sum_{I=1}^{N} \omega_I d^I_j \right)^2} \\
&= \frac{(2\pi kT)^{n/2}}{(2\pi)^N} \left(\prod_{i=1}^{n} m_i \right)^{1/2} \int_{-\infty}^{\infty} d\boldsymbol{\omega}\; e^{-i \sum_{I=1}^{N} \omega_I P_{I,x}} F^{\ell}_j(\boldsymbol{\omega}) \exp\left[-\frac{1}{2\beta} \sum_{i=1}^{n} m_i \left(\sum_{I=1}^{N} \omega_I d^I_i \right)^2 \right] \\
&= \frac{(2\pi kT)^{n/2}}{(2\pi)^N} \left(\prod_{i=1}^{n} m_i \right)^{1/2} \int_{-\infty}^{\infty} d\boldsymbol{\omega}\; e^{-i \sum_{I=1}^{N} \omega_I P_{I,x}} F^{\ell}_j(\boldsymbol{\omega}) e^{-\frac{1}{2\beta} \sum_{I,J=1}^{N} \omega_I \omega_J M_{IJ}},
\end{aligned} \tag{A4}
$$

in which

$$
F^{\ell}_j(\boldsymbol{\omega}) = \begin{cases} 1, & \ell = 0, \\ i \frac{m_j}{\beta} \sum_{I=1}^{N} \omega_I d^I_j, & \ell = 1, \\ \frac{m_i}{\beta} \left[1 - \frac{1}{\beta} \sum_{I,J=1}^{N} \omega_I \omega_J d^I_j d^J_j m_j \right], & \ell = 2, \end{cases} \tag{A5}
$$

and in the last line it is noted that $\sum_{i=1}^{n} m_i (\sum_{I=1}^{N} \omega_I d^I_i)^2 = \sum_{I=1}^{N} \sum_{J=1}^{N} \omega_I \omega_J \sum_{i=1}^{n} d^I_i d^J_i m_i = \sum_{I,J=1}^{N} \omega_I \omega_J M_{IJ}$, using the definition from Eq. (23). In fact, this last line is the motivation for the definition in Eq. (23) because we want to be able to ultimately extract this term from the integral. Since the CG variables are held fixed in the integration process, functions corresponding to these variables can be taken outside the integral. Note specifically that $M_I = \sum_{i=1}^{n} m_i$ does not appear naturally in $I^{\ell}_{j,x}$ and in fact no simplifications result using this expression. This clearly indicates that CG particle masses are not the preferred variables but the more general M_{IJ} are.

The remaining integrals over $\boldsymbol{\omega}$ are complicated by the fact that the exponential terms mix values for different I and J. This can be simplified by transforming to a new basis that diagonalizes the exponential factors. Consider $\boldsymbol{\omega}$ as an $N \times 1$ column matrix with elements ω_I so that $\sum_{I,J=1}^{N} \omega_I M_{IJ} \omega_J = \boldsymbol{\omega}^{\mathrm{T}} \mathbf{M} \boldsymbol{\omega}$ taking M_{IJ} as the elements of the mass matrix. Let $\mathbf{Q}$ be a matrix that transforms $\mathbf{M}$ to the basis of its eigenvectors, that is, $\mathbf{M} = \mathbf{Q}\mathbf{M}'\mathbf{Q}^{\mathrm{T}}$ with $\mathbf{M}'$ a diagonal matrix. Since $\mathbf{M}$ is symmetric, $\mathbf{Q}$ is unitary. This then gives $\boldsymbol{\omega}^{\mathrm{T}} \mathbf{M} \boldsymbol{\omega} = \boldsymbol{\omega}'^{\mathrm{T}} \mathbf{M}' \boldsymbol{\omega}' = \sum_{I=1}^{n} \omega'^2_I \lambda_I$ with $\boldsymbol{\omega}' = \mathbf{Q}^{\mathrm{T}} \boldsymbol{\omega}$ and λ_I the eigenvalues of $\mathbf{M}$. With this transformation, $\sum_{I=1}^{N} \omega_I P_{I,x} = \boldsymbol{\omega}^{\mathrm{T}} \mathbf{P}_x = \boldsymbol{\omega}'^{\mathrm{T}} \mathbf{P}'_x$ with $\mathbf{P}'_x = \mathbf{Q}^{\mathrm{T}} \mathbf{P}_x$, and the values of $F^{\ell}_j(\boldsymbol{\omega})$ are easily converted except for one term involving $\ell = 2$. Defining $\mathcal{M}^j_{IJ} = d^I_j d^J_j m_j$ then gives $\sum_{I,J=1}^{N} \omega_I \omega_J \mathcal{M}^j_{IJ} = \boldsymbol{\omega}^{\mathrm{T}} \boldsymbol{\mathcal{M}}^j \boldsymbol{\omega} = \boldsymbol{\omega}'^{\mathrm{T}} \boldsymbol{\mathcal{M}}^{j\prime} \boldsymbol{\omega}'$ with $\boldsymbol{\mathcal{M}}^{j\prime} = \mathbf{Q}^{\mathrm{T}} \boldsymbol{\mathcal{M}}^j \mathbf{Q}$. Note that $\sum_{j=1}^{n} \mathcal{M}^j_{IJ} = M_{IJ}$. Finally, the Jacobian determinant for the transformation is $|\mathbf{Q}| = 1$. Putting all these pieces together allows the integral in $I^{\ell}_{j,x}$ to be transformed as

$$
\begin{aligned}
I^{\ell}_{j,x} &= \frac{(2\pi kT)^{n/2}}{(2\pi)^N} \left(\prod_{i=1}^{n} m_i \right)^{1/2} \int_{-\infty}^{\infty} d\boldsymbol{\omega}'\; e^{-i \sum_{I=1}^{N} \omega'_I P'_{I,x}} \\
&\quad \times F^{\ell}_j(\boldsymbol{\omega}') e^{-\frac{1}{2\beta} \sum_{I=1}^{N} \omega'^2_I \lambda_I},
\end{aligned} \tag{A6}
$$

in which terms can now be factored and integrals can be performed by completing the square in the exponential term. The case for $\ell = 0$ gives

$$
\begin{aligned}
I^0_{j,x} = I^0_x &= \frac{(2\pi kT)^{n/2}}{(2\pi)^N} \left(\prod_{i=1}^{n} m_i \right)^{1/2} \left\{ \prod_{I=1}^{N} \int_{-\infty}^{\infty} d\omega'_I\; e^{-i\omega'_I P'_{I,x}} e^{-\frac{1}{2\beta} \omega'^2_I \lambda_I} \right\} \\
&= \frac{(2\pi kT)^{n/2}}{(2\pi)^N} \left(\prod_{i=1}^{n} m_i \right)^{1/2} \left\{ \prod_{I=1}^{N} \sqrt{\frac{2\pi\beta}{\lambda_I}} e^{-\frac{\beta P'^2_{I,x}}{2\lambda_I}} \right\} \\
&= (2\pi kT)^{(n-N)/2} \left(\prod_{i=1}^{n} m_i \right)^{1/2} \left(\prod_{I=1}^{N} \lambda_I \right)^{-1/2} \exp\left[-\frac{\beta}{2} \sum_{I=1}^{N} \frac{P'^2_{I,x}}{\lambda_I} \right] \\
&= (2\pi kT)^{(n-N)/2} \left(\prod_{i=1}^{n} m_i \right)^{1/2} |\mathbf{M}|^{-1/2} \exp\left[-\frac{\beta}{2} \sum_{I,J=1}^{N} P_{I,x} M^{-1}_{IJ} P_{J,x} \right],
\end{aligned} \tag{A7}
$$

in which the last summations arise from

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$$\sum_{I=1}^{N}\frac{P'^2_{I,x}}{\lambda_I}=\mathbf{P}'^{T}_x\mathbf{M}'^{-1}\mathbf{P}'_x=(\mathbf{Q}^T\mathbf{P}_x)^T(\mathbf{Q}^T\mathbf{M}\mathbf{Q})^{-1}(\mathbf{Q}^T\mathbf{P}_x)=\mathbf{P}_x^T\mathbf{M}^{-1}\mathbf{P}_x. \tag{A8}$$

It should be noted that the results above hold only if none of the eigenvalues of **M** are zero. If for some particular J, $\lambda_J = 0$ then the corresponding integral over ω'_J produces $\delta[P'_{J,x}]$ which eliminates all contributions to the projection except those where $P'_{J,x} = 0$. This forces a constraint among the values of $P_{I,x}$ whose impact on the equations of motion must then be determined. This will not be done here because for many physical systems, it is anticipated that all the eigenvalues of **M** will be non-zero.

The case for $\ell = 1$ gives

$$\begin{aligned}
I^1_{j,x}&=\frac{(2\pi kT)^{n/2}}{(2\pi)^N}\left(\prod_{i=1}^{n}m_i\right)^{1/2}\int_{-\infty}^{\infty}d\boldsymbol{\omega}'\left[\frac{im_j}{\beta}\sum_{J,K=1}^{N}Q_{JK}\omega'_K d^J_j\right]e^{-i\sum_{I=1}^{N}\omega'_I P'_{I,x}}e^{-\frac{1}{2\beta}\sum_{I=1}^{N}\omega'^2_I\lambda_I}\\
&=i\frac{m_j(2\pi kT)^{n/2}}{\beta(2\pi)^N}\left(\prod_{i=1}^{n}m_i\right)^{1/2}\left[\sum_{J,K=1}^{N}Q_{JK}d^J_j\int_{-\infty}^{\infty}d\omega'_K\,\omega'_K e^{-i\omega'_K P'_{K,x}}e^{-\frac{1}{2\beta}\omega'^2_K\lambda_K}\left\{\prod_{I\neq K}\int_{-\infty}^{\infty}d\omega'_I\,e^{-i\omega'_I P'_{I,x}}e^{-\frac{1}{2\beta}\omega'^2_I\lambda_I}\right\}\right]\\
&=i\frac{m_j(2\pi kT)^{n/2}}{\beta(2\pi)^N}\left(\prod_{i=1}^{n}m_i\right)^{1/2}\left[\sum_{J,K=1}^{N}Q_{JK}d^J_j\left\{-i\frac{\beta}{\lambda_K}P'_{K,x}\right\}\sqrt{\frac{2\pi\beta}{\lambda_K}}e^{-\frac{\beta P'^2_{K,x}}{2\lambda_K}}\left\{\prod_{I\neq K}\sqrt{\frac{2\pi\beta}{\lambda_I}}e^{-\frac{\beta P'^2_{I,x}}{2\lambda_I}}\right\}\right]\\
&=I^0_x\sum_{J=1}^{N}d^J_j m_j\sum_{K,L=1}^{N}Q_{JK}\frac{1}{\lambda_K}Q^T_{KL}P_{L,x}=I^0_x\sum_{I,J=1}^{N}d^I_j m_j M^{-1}_{IJ}P_{J,x}.
\end{aligned} \tag{A9}$$

For $\ell = 2$, some care is needed to correctly account for the various terms, that is,

$$\begin{aligned}
I^2_{j,x}&=\frac{(2\pi kT)^{n/2}}{(2\pi)^N}\left(\prod_{i=1}^{n}m_i\right)^{1/2}\int_{-\infty}^{\infty}d\boldsymbol{\omega}'\,\frac{m_j}{\beta}\left[1-\frac{1}{\beta}\sum_{K,L=1}^{N}\omega'_K\omega'_L\mathcal{M}'^{j'}_{KL}\right]e^{-i\sum_{I=1}^{N}\omega'_I P'_{I,x}}e^{-\frac{1}{2\beta}\sum_{I=1}^{N}\omega'^2_I\lambda_I}\\
&=m_jkTI^0_x-\frac{m_j(2\pi kT)^{n/2}}{\beta^2(2\pi)^N}\left(\prod_{i=1}^{n}m_i\right)^{1/2}\left\{\sum_{K=1}^{N}\mathcal{M}'^{j'}_{KK}\int_{-\infty}^{\infty}d\boldsymbol{\omega}'\,\omega'^2_K e^{-i\sum_{I=1}^{N}\omega'_I P'_{I,x}}e^{-\frac{1}{2\beta}\sum_{I=1}^{N}\omega'^2_I\lambda_I}\right.\\
&\quad\left.+2\sum_{K=2}^{N}\sum_{L=1}^{K-1}\mathcal{M}'^{j'}_{KL}\int_{-\infty}^{\infty}d\boldsymbol{\omega}'\,\omega'_K\omega'_L e^{-i\sum_{I=1}^{N}\omega'_I P'_{I,x}}e^{-\frac{1}{2\beta}\sum_{I=1}^{N}\omega'^2_I\lambda_I}\right\}\\
&=m_jkTI^0_x-\frac{m_j(2\pi kT)^{n/2}}{\beta^2(2\pi)^N}\left(\prod_{i=1}^{n}m_i\right)^{1/2}\left[\sum_{K=1}^{N}\mathcal{M}'^{j'}_{KK}\int_{-\infty}^{\infty}d\omega'_K\,\omega'^2_K e^{-i\omega'_K P'_{K,x}}e^{-\frac{1}{2\beta}\omega'^2_K\lambda_K}\left\{\prod_{I\neq K}\int_{-\infty}^{\infty}d\omega'_I\,e^{-i\omega'_I P'_{I,x}}e^{-\frac{1}{2\beta}\omega'^2_I\lambda_I}\right\}\right.\\
&\quad\left.+2\sum_{K=2}^{N}\sum_{L=1}^{K-1}\mathcal{M}'^{j'}_{KL}\int_{-\infty}^{\infty}d\omega'_K\,\omega'_K e^{-i\omega'_K P'_{K,x}}e^{-\frac{1}{2\beta}\omega'^2_K\lambda_K}\int_{-\infty}^{\infty}d\omega'_L\,\omega'_L e^{-i\omega'_L P'_{L,x}}e^{-\frac{1}{2\beta}\omega'^2_L\lambda_L}\left\{\prod_{I\neq K,L}\int_{-\infty}^{\infty}d\omega'_I\,e^{-i\omega'_I P'_{I,x}}e^{-\frac{1}{2\beta}\omega'^2_I\lambda_I}\right\}\right]\\
&=m_jkTI^0_x-\frac{m_j(2\pi kT)^{n/2}}{\beta^2(2\pi)^N}\left(\prod_{i=1}^{n}m_i\right)^{1/2}\left[\sum_{K=1}^{N}\mathcal{M}'^{j'}_{KK}\left\{\frac{\beta}{\lambda_K}-\frac{\beta^2}{\lambda_K^2}P'^2_{K,x}\right\}\sqrt{\frac{2\pi\beta}{\lambda_K}}e^{-\frac{\beta P'^2_{K,x}}{2\lambda_K}}\left\{\prod_{I\neq K}\sqrt{\frac{2\pi\beta}{\lambda_I}}e^{-\frac{\beta P'^2_{I,x}}{2\lambda_I}}\right\}\right.\\
&\quad\left.+2\sum_{K=2}^{N}\sum_{L=1}^{K-1}\mathcal{M}'^{j'}_{KL}\left\{\frac{\beta P'_{K,x}}{i\lambda_K}\sqrt{\frac{2\pi\beta}{\lambda_K}}e^{-\frac{\beta P'^2_{K,x}}{2\lambda_K}}\right\}\left\{\frac{\beta P'_{L,x}}{i\lambda_L}\sqrt{\frac{2\pi\beta}{\lambda_L}}e^{-\frac{\beta P'^2_{L,x}}{2\lambda_L}}\right\}\left\{\prod_{I\neq K,L}\sqrt{\frac{2\pi\beta}{\lambda_I}}e^{-\frac{\beta P'^2_{I,x}}{2\lambda_I}}\right\}\right]\\
&=m_jkTI^0_x-m_jkTI^0_x\sum_{K=1}^{N}\frac{\mathcal{M}'^{j'}_{KK}}{\lambda_K}+m_jI^0_x\sum_{K,L=1}^{N}\frac{P'_{K,x}}{\lambda_K}\mathcal{M}'^{j'}_{KL}\frac{P'_{L,x}}{\lambda_L}\\
&=m_jkTI^0_x\left[1-\sum_{K=1}^{N}\sum_{I,I'=1}^{N}Q^T_{KI}\frac{\mathcal{M}^j_{II'}}{\lambda_K}Q_{I'K}+\beta\sum_{K,L=1}^{N}\sum_{I,I'=1}^{N}\sum_{J,J'=1}^{N}\frac{P_{J,x}}{\lambda_K}Q_{JK}Q^T_{KI}\mathcal{M}^j_{II'}Q_{I'L}Q^T_{LJ'}\frac{P_{J',x}}{\lambda_L}\right]\\
&=m_jkTI^0_x\left[1-\sum_{I,I'=1}^{N}d^I_j d^{I'}_j m_j M^{-1}_{I'I}+\beta\sum_{I,I'=1}^{N}\sum_{J,J'=1}^{N}P_{J,x}M^{-1}_{JI}d^I_j d^{I'}_j m_j M^{-1}_{I'J'}P_{J',x}\right].
\end{aligned} \tag{A10}$$

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Because all components are treated equivalently, analogous results for $I^{\ell}_{j,y}$ and $I^{\ell}_{j,z}$ are obtained by replacing all occurrences of x in the expressions for $I^{\ell}_{j,x}$ with y or z, respectively.

These results can now be used to evaluate expressions for the projection operator acting on various functions. We begin with the entropy, starting with the evaluation of $\Omega(\mathbf{W}, \mathbf{P}, \mathbf{M})$ using Eq. (A1), that is,

$$\begin{aligned}\Omega(\mathbf{W},\mathbf{P},\mathbf{M}) &= \frac{1}{Z}\int d\mathbf{r}\; e^{-\beta U(\mathbf{r})}\left\{\prod_{I=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I m_i(\mathbf{r}_i - \mathbf{R}^I_{lab})\right) - \mathbf{W}_I\right]\right\}\left\{\prod_{I,J=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I d_i^J m_i\right) - \mathrm{M}_{IJ}\right]\right\} \times \mathrm{I}^0_x \times \mathrm{I}^0_y \times \mathrm{I}^0_z \\ &= \frac{1}{Z}\int d\mathbf{r}\; e^{-\beta U(\mathbf{r})}\left\{\prod_{I=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I m_i(\mathbf{r}_i - \mathbf{R}^I_{lab})\right) - \mathbf{W}_I\right]\right\}\left\{\prod_{I,J=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I d_i^J m_i\right) - \mathrm{M}_{IJ}\right]\right\}(2\pi kT)^{3(n-N)/2}\left(\prod_{i=1}^{n} m_i\right)^{3/2}|\mathbf{M}|^{-3/2} \\ &\quad\times \exp\left[-\frac{\beta}{2}\sum_{I,J=1}^{N}\left(\mathrm{P}_{I,x}\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,x} + \mathrm{P}_{I,y}\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,y} + \mathrm{P}_{I,z}\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,z}\right)\right] \\ &= (2\pi kT)^{-3N/2}|\mathbf{M}|^{-3/2}\exp\left[-\beta \mathrm{V}(\mathbf{W},\mathbf{M}) - \frac{\beta}{2}\sum_{I,J=1}^{N}\left(\mathrm{P}_{I,x}\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,x} + \mathrm{P}_{I,y}\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,y} + \mathrm{P}_{I,z}\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,z}\right)\right],\end{aligned} \tag{A11}$$

with V $(\mathbf{W}, \mathbf{M})$ defined in Eq. (27). The delta functions within the integrals ensure that the values of the CG variables remain fixed during the integrations. This allows all factors of $\mathbf{W}_I$, $\mathbf{P}_I$, and M_{IJ} to be taken outside the integral and again demonstrates the importance of defining the CG mass variables M_{IJ} to enable this. The expression for the entropy in Eq. (30) is obtained by substituting the expression above into Eq. (6).

A short departure shows that if the mapping of Eq. (10) along with the choice of Eq. (13) is used as the starting point, and the derivation above followed, the last line of Eq. (A4) is the same except the exponent involves $\sum_{i=1}^{n} m_i(\sum_{I=1}^{N}\omega_I c_i^I d_i^I)^2 = \sum_{I=1}^{N}\sum_{J=1}^{N}\omega_I\omega_J\sum_{n=1}^{n} m_i c_i^I c_i^J d_i^I d_i^J = \sum_{I=1}^{N}\omega_I^2\widetilde{\mathrm{M}}_I$ with $\widetilde{\mathrm{M}}_I = \sum_{i=1}^{n}(c_i^I d_i^I)^2 m_i$. This follows directly from Eq. (13) since an atomistic particle can be part of only a single CG particle, it must be that $d_i^I d_i^J = (d_i^I)^2\delta_{IJ}$. The exponent becomes a pure quadratic function of ω_I and since $\widetilde{\mathrm{M}}_I$ is constant in this case, the integral over the ω space can be performed as is giving $\mathrm{I}^0_x = (2\pi kT)^{(n-N)/2}(\prod_{i=1}^{n} m_i/\prod_{I=1}^{N}\mathrm{M}_I)^{1/2}\exp[-\beta/2\sum_{I=1}^{N}\mathrm{P}^2_{I,x}/\widetilde{\mathrm{M}}_I]$. Similar expressions result for I^0_y and I^0_z which when used in the equation above ultimately lead to the expression for $\Omega(\mathbf{R}, \mathbf{P})$ given by Eq. (17).

To complete the equations of motion, the result of P_{α} acting on $\mathbf{\Lambda}^1_{\mathrm{K}}$, $\mathbf{\Lambda}^2_{\mathrm{K}}$, $\mathbf{\Lambda}^3_{\mathrm{KL}}$, and $\mathbf{F}^1_{\mathrm{K}}$ must be determined. The derivation of these results is less cluttered if the momentum part of $\mathcal{P}$ (here denoted $\mathcal{P}_{\mathbf{p}}$) is evaluated first so that [using the definitions from Eqs. (25)]

$$\begin{aligned}\mathcal{P}_{\mathbf{p}}(\mathbf{\Lambda}^1_{\mathrm{K}}) &= \int_{-\infty}^{\infty} d\mathbf{p}\left[\sum_{j=1}^{n}\left(\mathbf{p}_j\cdot\frac{\partial d_j^{\mathrm{K}}}{\partial \mathbf{r}_j}\right)(\mathbf{r}_j - \mathbf{R}^{\mathrm{K}}_{lab})\right]e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{I=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I\mathbf{p}_i\right) - \mathbf{P}_I\right]\right\} \\ &= \sum_{j=1}^{n}(\mathbf{r}_j - \mathbf{R}^{\mathrm{K}}_{lab})\sum_{\alpha=x,y,z}\frac{\partial d_j^{\mathrm{K}}}{\partial r_{j,\alpha}}\int_{-\infty}^{\infty} d\mathbf{p}\; p_{j,\alpha}e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{I=1}^{N}\delta\left[\left(\sum_{i=1}^{n} d_i^I\mathbf{p}_i\right) - \mathbf{P}_I\right]\right\} \\ &= \sum_{j=1}^{n}(\mathbf{r}_j - \mathbf{R}^{\mathrm{K}}_{lab})\sum_{\alpha=x,y,z}\frac{\partial d_j^{\mathrm{K}}}{\partial r_{j,\alpha}}\times \mathrm{I}^1_{j,\alpha}\times \mathrm{I}^0_{\beta}\times \mathrm{I}^0_{\gamma} \\ &= \mathrm{I}^0_x\mathrm{I}^0_y\mathrm{I}^0_z\sum_{j=1}^{n}(\mathbf{r}_j - \mathbf{R}^{\mathrm{K}}_{lab})\sum_{\alpha=x,y,z}\frac{\partial d_j^{\mathrm{K}}}{\partial r_{j,\alpha}}\sum_{I,J=1}^{N} m_j d_j^I\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,\alpha} \\ &= \mathrm{I}^0_x\mathrm{I}^0_y\mathrm{I}^0_z\sum_{\alpha=x,y,z}\sum_{I,J=1}^{N}(\mathbf{G}^1_{\mathrm{KI}})_{\alpha}\mathrm{M}^{-1}_{IJ}\mathrm{P}_{J,\alpha},\end{aligned} \tag{A12}$$

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in which the subscripts β and γ take on the values of the components other than α, and $\mathbf{G}^1_{\mathrm{KI}}$ is defined by Eq. (33). Adding the result above into the full integral gives

$$\mathrm{P}_{\boldsymbol{\alpha}}\boldsymbol{\Lambda}^1_{\mathrm{K}} = \langle\boldsymbol{\Lambda}^1_{\mathrm{K}}\rangle = \sum_{\mathrm{I,J}=1}^{\mathrm{N}}\langle\mathbf{G}^1_{\mathrm{KI}}\rangle\cdot\mathrm{M}^{-1}_{\mathrm{IJ}}\mathbf{P}_{\mathrm{J}}, \tag{A13}$$

with

$$\langle\mathbf{G}^1_{\mathrm{KI}}\rangle = \frac{1}{\mathrm{Z}_r}\int d\mathbf{r}\; e^{-\beta[\mathrm{U}(\mathbf{r})-\mathrm{V}(\mathbf{W},\mathbf{M})]}\left(\sum_{i=1}^{n} m_i d^{\mathrm{I}}_i m_i(\mathbf{r}_i-\mathbf{R}^{\mathrm{K}}_{lab})\frac{\partial d^{\mathrm{K}}_i}{\partial\mathbf{r}_i}\right)\left\{\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i m_i(\mathbf{r}_i-\mathbf{R}^{\mathrm{I}}_{lab})\right)-\mathbf{W}_{\mathrm{I}}\right]\right\}\left\{\prod_{\mathrm{I,J}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i d^{\mathrm{J}}_i m_i\right)-\mathrm{M}_{\mathrm{IJ}}\right]\right\}. \tag{A14}$$

An analogous procedure for $\boldsymbol{\Lambda}^3_{\mathrm{KL}}$ gives

$$\begin{aligned}
\mathcal{P}_{\mathbf{p}}(\boldsymbol{\Lambda}^3_{\mathrm{KL}}) &= \int_{-\infty}^{\infty} d\mathbf{p}\left[\sum_{j=1}^{n}\mathbf{p}_j\cdot\left(d^{\mathrm{L}}_j\frac{\partial d^{\mathrm{K}}_j}{\partial\mathbf{r}_j}+d^{\mathrm{K}}_j\frac{\partial d^{\mathrm{L}}_j}{\partial\mathbf{r}_j}\right)\right]e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i\mathbf{p}_i\right)-\mathbf{P}_I\right]\right\}\\
&= \sum_{j=1}^{n}\sum_{\alpha=x,y,z}\left(d^{\mathrm{L}}_j\frac{\partial d^{\mathrm{K}}_j}{\partial r_{j,\alpha}}+d^{\mathrm{K}}_j\frac{\partial d^{\mathrm{L}}_j}{\partial r_{j,\alpha}}\right)\int_{-\infty}^{\infty}d\mathbf{p}\; p_{j,\alpha}e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i\mathbf{p}_i\right)-\mathbf{P}_I\right]\right\}\\
&= \mathrm{I}^0_x\mathrm{I}^0_y\mathrm{I}^0_z\sum_{j=1}^{n}\sum_{\alpha=x,y,z}\left(d^{\mathrm{L}}_j\frac{\partial d^{\mathrm{K}}_j}{\partial r_{j,\alpha}}+d^{\mathrm{K}}_j\frac{\partial d^{\mathrm{L}}_j}{\partial r_{j,\alpha}}\right)\sum_{\mathrm{I,J}=1}^{\mathrm{N}} m_j d^{\mathrm{I}}_j\mathrm{M}^{-1}_{\mathrm{IJ}}\mathrm{P}_{\mathrm{J},\alpha}\\
&= \mathrm{I}^0_x\mathrm{I}^0_y\mathrm{I}^0_z\sum_{\alpha=x,y,z}\sum_{\mathrm{I,J}=1}^{\mathrm{N}}[(\mathbf{G}^2_{\mathrm{ILK}})_\alpha+(\mathbf{G}^2_{\mathrm{IKL}})_\alpha]\mathrm{M}^{-1}_{\mathrm{IJ}}\mathrm{P}_{\mathrm{J},\alpha},
\end{aligned} \tag{A15}$$

with $\mathbf{G}^2_{\mathrm{IKL}}$ defined by Eq. (34) so that

$$\mathrm{P}_{\boldsymbol{\alpha}}\boldsymbol{\Lambda}^3_{\mathrm{KL}} = \langle\boldsymbol{\Lambda}^3_{\mathrm{KL}}\rangle = \sum_{\mathrm{I,J}=1}^{\mathrm{N}}[\langle\mathbf{G}^2_{\mathrm{ILK}}\rangle+\langle\mathbf{G}^2_{\mathrm{ILK}}\rangle]\cdot\mathrm{M}^{-1}_{\mathrm{IJ}}\mathbf{P}_{\mathrm{J}}, \tag{A16}$$

with

$$\langle\mathbf{G}^2_{\mathrm{IJK}}\rangle = \frac{1}{\mathrm{Z}_r}\int d\mathbf{r}\; e^{-\beta[\mathrm{U}(\mathbf{r})-\mathrm{V}(\mathbf{W},\mathbf{M})]}\left(\sum_{i=1}^{n} m_i d^{\mathrm{I}}_i d^{\mathrm{J}}_i\frac{\partial d^{\mathrm{K}}_i}{\partial\mathbf{r}_i}\right)\left\{\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i m_i(\mathbf{r}_i-\mathbf{R}^{\mathrm{I}}_{lab})\right)-\mathbf{W}_{\mathrm{I}}\right]\right\}\left\{\prod_{\mathrm{I,J}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i d^{\mathrm{J}}_i m_i\right)-\mathrm{M}_{\mathrm{IJ}}\right]\right\}. \tag{A17}$$

The procedure is a bit more involved for $\boldsymbol{\Lambda}^2_{\mathrm{K}}$, namely,

$$\begin{aligned}
\mathcal{P}_{\mathbf{p}}(\boldsymbol{\Lambda}^2_{\mathrm{K}}) &= \int_{-\infty}^{\infty} d\mathbf{p}\left[\sum_{j=1}^{n}\left(\frac{\mathbf{p}_j}{m_j}\cdot\frac{\partial d^{\mathrm{K}}_j}{\partial\mathbf{r}_j}\right)\mathbf{p}_j\right]e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i\mathbf{p}_i\right)-\mathbf{P}_I\right]\right\}\\
&= \sum_{j=1}^{n}\frac{1}{m_j}\left[\sum_{\alpha=x,y,z}\frac{\partial d^{\mathrm{K}}_j}{\partial r_{j,\alpha}}\hat{\alpha}\int_{-\infty}^{\infty}d\mathbf{p}\; p^2_{j,\alpha}e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i\mathbf{p}_i\right)-\mathbf{P}_I\right]\right\}\right.\\
&\quad\left.+\sum_{\alpha\neq\beta}\frac{\partial d^{\mathrm{K}}_j}{\partial r_{j,\beta}}\hat{\alpha}\int_{-\infty}^{\infty}d\mathbf{p}\; p_{j,\beta}p_{j,\alpha}e^{-\beta\sum_{i=1}^{n}\frac{p_i^2}{2m_i}}\left\{\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{i=1}^{n} d^{\mathrm{I}}_i\mathbf{p}_i\right)-\mathbf{P}_I\right]\right\}\right]\\
&= \mathrm{I}^0_x\mathrm{I}^0_y\mathrm{I}^0_z\sum_{j=1}^{n}\frac{1}{m_j}\left\{\sum_{\alpha=x,y,z}\frac{\partial d^{\mathrm{K}}_j}{\partial r_{j,\alpha}}\hat{\alpha}\times(m_jkT)\left[1-\sum_{\mathrm{I,I'}=1}^{\mathrm{N}} m_j d^{\mathrm{I}}_j d^{\mathrm{I'}}_j\mathrm{M}^{-1}_{\mathrm{I'I}}+\beta\sum_{\mathrm{I,I'}=1}^{\mathrm{N}}\sum_{\mathrm{J,J'}=1}^{\mathrm{N}}\mathrm{P}_{\mathrm{J},\alpha}\mathrm{M}^{-1}_{\mathrm{JI}}m_j d^{\mathrm{I}}_j d^{\mathrm{I'}}_j\mathrm{M}^{-1}_{\mathrm{I'J'}}\mathrm{P}_{\mathrm{J'},\alpha}\right]\right.\\
&\quad\left.+\sum_{\alpha\neq\beta}\frac{\partial d^{\mathrm{K}}_j}{\partial r_{j,\beta}}\hat{\alpha}\left(\sum_{\mathrm{I,J}=1}^{\mathrm{N}} m_j d^{\mathrm{I}}_j\mathrm{M}^{-1}_{\mathrm{IJ}}\mathrm{P}_{\mathrm{J},\beta}\right)\left(\sum_{\mathrm{I'J'}=1}^{\mathrm{N}} m_j d^{\mathrm{I'}}_j\mathrm{M}^{-1}_{\mathrm{I'J'}}\mathrm{P}_{\mathrm{J'},\alpha}\right)\right\}\\
&= \mathrm{I}^0_x\mathrm{I}^0_y\mathrm{I}^0_z\left\{kT\sum_{j=1}^{n}\frac{\partial d^{\mathrm{K}}_j}{\partial\mathbf{r}_j}-kT\sum_{\mathrm{I,I'}=1}^{\mathrm{N}}\mathbf{G}^2_{\mathrm{II'K}}\mathrm{M}^{-1}_{\mathrm{I'I}}+\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\sum_{\mathrm{I',J'}=1}^{\mathrm{N}}\left(\sum_{\beta=x,y,z}(\mathbf{G}^2_{\mathrm{II'K}})_\beta\mathrm{M}^{-1}_{\mathrm{IJ}}\mathrm{P}_{\mathrm{J},\beta}\right)\left(\sum_{\alpha=x,y,z}\mathrm{M}^{-1}_{\mathrm{I'J'}}\mathrm{P}_{\mathrm{J'},\alpha}\hat{\alpha}\right)\right\},
\end{aligned} \tag{A18}$$

in which $\hat{\alpha}$ is a unit vector in the Cartesian direction α and where it is noted that the inverse of the symmetric matrix $\mathbf{M}$ is also symmetric. Adding this result into the full integral then gives

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$$\mathrm{P}_{\boldsymbol{\alpha}}\boldsymbol{\Lambda}_{\mathrm{K}}^{2}=\langle\boldsymbol{\Lambda}_{\mathrm{K}}^{2}\rangle=kT\langle\sum_{\mathrm{j}=1}^{\mathrm{n}}\frac{\partial d_{\mathrm{j}}^{\mathrm{K}}}{\partial\mathbf{r}_{\mathrm{j}}}\rangle-kT\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\langle\mathbf{G}_{\mathrm{IJK}}^{2}\rangle\mathrm{M}_{\mathrm{IJ}}^{-1}+\sum_{\mathrm{I',J'}=1}^{\mathrm{N}}\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\left[\langle\mathbf{G}_{\mathrm{II'K}}^{2}\rangle\cdot\mathrm{M}_{\mathrm{IJ}}^{-1}\mathbf{P}_{\mathrm{J}}\right]\mathrm{M}_{\mathrm{I'J'}}^{-1}\mathbf{P}_{\mathrm{J'}}. \tag{A19}$$

Only $\mathrm{P}_{\boldsymbol{\alpha}}\mathbf{F}_{\mathrm{K}}$ remains to be determined and this requires a slightly different approach. Since U is considered to be a function of positions only, the momentum part of $\mathcal{P}\mathbf{F}_{\mathrm{K}}$ is just $\mathrm{I}_{\mathrm{x}}^{0}\mathrm{I}_{\mathrm{y}}^{0}\mathrm{I}_{\mathrm{z}}^{0}$ so that

$$\mathcal{P}\mathbf{F}_{\mathrm{K}}=-\frac{\mathrm{I}_{\mathrm{x}}^{0}\mathrm{I}_{\mathrm{y}}^{0}\mathrm{I}_{\mathrm{z}}^{0}}{\mathrm{Z}}\sum_{\mathrm{j}=1}^{\mathrm{n}}\sum_{\alpha=\mathrm{x,y,z}}\hat{\alpha}\int d\mathbf{r}\,d_{\mathrm{j}}^{\mathrm{K}}\frac{\partial\mathrm{U}}{\partial r_{\mathrm{j},\alpha}}e^{-\beta\mathrm{U}(\mathbf{r})}\left\{\prod_{\beta=\mathrm{x,y,z}}\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}m_{\mathrm{i}}(r_{\mathrm{i},\beta}-\mathrm{R}_{\mathrm{lab},\beta}^{\mathrm{I}})\right)-\mathrm{W}_{\mathrm{I},\beta}\right]\right\}\left\{\prod_{\mathrm{I,J}=1}^{\mathrm{N}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}d_{\mathrm{i}}^{\mathrm{J}}m_{\mathrm{i}}\right)-\mathrm{M}_{\mathrm{IJ}}\right]\right\}. \tag{A20}$$

Recognizing that $(\partial\mathrm{U}/\partial r_{\mathrm{j},\alpha})\exp(-\beta\mathrm{U})=-kT\,(\partial\exp(-\beta\mathrm{U})/\partial r_{\mathrm{j},\alpha})$, then integrating by parts (assuming the potential decays to zero quickly enough as $r_{\mathrm{j},\alpha}\to\pm\infty$), and employing the delta function relation $d\delta[x-a]/dx=-d\delta[x-a]/da$ allows $\mathcal{P}\mathbf{F}_{\mathrm{K}}$ to be written as

$$\begin{aligned}
\mathcal{P}\mathbf{F}_{\mathrm{K}}=&-\frac{\mathrm{I}_{\mathrm{x}}^{0}\mathrm{I}_{\mathrm{y}}^{0}\mathrm{I}_{\mathrm{z}}^{0}}{\beta\mathrm{Z}}\sum_{\mathrm{j}=1}^{\mathrm{n}}\sum_{\alpha=\mathrm{x,y,z}}\hat{\alpha}\int d\mathbf{r}\,e^{-\beta\mathrm{U}(\mathbf{r})}\Bigg[\frac{\partial d_{\mathrm{j}}^{\mathrm{K}}}{\partial r_{\mathrm{j},\alpha}}\left\{\prod_{\beta=\mathrm{x,y,z}}\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}m_{\mathrm{i}}(r_{\mathrm{i},\beta}-\mathrm{R}_{\mathrm{lab},\beta}^{\mathrm{I}})\right)-\mathrm{W}_{\mathrm{I},\beta}\right]\right\}\\
&\times\left\{\prod_{\mathrm{I,J}=1}^{\mathrm{N}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}d_{\mathrm{i}}^{\mathrm{J}}m_{\mathrm{i}}\right)-\mathrm{M}_{\mathrm{IJ}}\right]\right\}-d_{\mathrm{j}}^{\mathrm{K}}\sum_{\mathrm{I}=1}^{\mathrm{N}}\sum_{\beta=\mathrm{x,y,z}}\left\{\frac{\partial}{\partial\mathrm{W}_{\mathrm{I},\beta}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}m_{\mathrm{i}}(r_{\mathrm{i},\beta}-\mathrm{R}_{\mathrm{lab},\beta}^{\mathrm{I}})\right)-\mathrm{W}_{\mathrm{I},\beta}\right]\right\}\\
&\times\left[\frac{\partial d_{\mathrm{j}}^{\mathrm{I}}}{\partial r_{\mathrm{j},\alpha}}m_{\mathrm{j}}(r_{\mathrm{j},\beta}-\mathrm{R}_{\mathrm{lab},\beta}^{\mathrm{I}})+d_{\mathrm{j}}^{\mathrm{I}}m_{\mathrm{j}}\delta_{\alpha\beta}\right]\left\{\prod_{\gamma\neq\beta}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}m_{\mathrm{i}}(r_{\mathrm{i},\gamma}-\mathrm{R}_{\mathrm{lab},\gamma}^{\mathrm{I}})\right)-\mathrm{W}_{\mathrm{I},\gamma}\right]\right\}\\
&\times\left\{\prod_{\gamma=\mathrm{x,y,z}}\prod_{\mathrm{J}\neq\mathrm{I}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{J}}m_{\mathrm{i}}(r_{\mathrm{i},\gamma}-\mathrm{R}_{\mathrm{lab},\gamma}^{\mathrm{J}})\right)-\mathrm{W}_{\mathrm{J},\gamma}\right]\right\}\left\{\prod_{\mathrm{I,J}=1}^{\mathrm{N}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}d_{\mathrm{i}}^{\mathrm{J}}m_{\mathrm{i}}\right)-\mathrm{M}_{\mathrm{IJ}}\right]\right\}\\
&-d_{\mathrm{j}}^{\mathrm{K}}\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\left\{\frac{\partial}{\partial\mathrm{M}_{\mathrm{IJ}}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}d_{\mathrm{i}}^{\mathrm{J}}m_{\mathrm{i}}\right)-\mathrm{M}_{\mathrm{IJ}}\right]\right\}\left[m_{\mathrm{j}}\frac{\partial d_{\mathrm{j}}^{\mathrm{I}}}{\partial r_{\mathrm{j},\alpha}}d_{\mathrm{j}}^{\mathrm{J}}+m_{\mathrm{j}}d_{\mathrm{j}}^{\mathrm{I}}\frac{\partial d_{\mathrm{j}}^{\mathrm{J}}}{\partial r_{\mathrm{j},\alpha}}\right]\\
&\times\left\{\prod_{\beta=\mathrm{x,y,z}}\prod_{\mathrm{I}=1}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}m_{\mathrm{i}}(r_{\mathrm{i},\beta}-\mathrm{R}_{\mathrm{lab},\beta}^{\mathrm{I}})\right)-\mathrm{W}_{\mathrm{I},\beta}\right]\right\}\left\{\prod_{\mathrm{K,L}\neq\mathrm{I,J}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{K}}d_{\mathrm{i}}^{\mathrm{L}}m_{\mathrm{i}}\right)-\mathrm{M}_{\mathrm{KL}}\right]\right\}\Bigg]\\
=&-\frac{\mathrm{I}_{\mathrm{x}}^{0}\mathrm{I}_{\mathrm{y}}^{0}\mathrm{I}_{\mathrm{z}}^{0}}{\beta\mathrm{Z}}\int d\mathbf{r}\,e^{-\beta\mathrm{U}(\mathbf{r})}\left\{\sum_{\mathrm{j}=1}^{\mathrm{n}}\frac{\partial d_{\mathrm{j}}^{\mathrm{K}}}{\partial\mathbf{r}_{\mathrm{j}}}-\sum_{\mathrm{I}=1}^{\mathrm{N}}\sum_{\alpha,\beta=\mathrm{x,y,z}}\hat{\alpha}\frac{\partial}{\partial\mathrm{W}_{\mathrm{I},\beta}}\left[(\mathbf{G}_{\mathrm{IK}}^{1})_{\beta\alpha}+\mathrm{M}_{\mathrm{KI}}\delta_{\alpha\beta}\right]-\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\sum_{\alpha=\mathrm{x,y,z}}\hat{\alpha}\frac{\partial}{\partial\mathrm{M}_{\mathrm{IJ}}}\left[(\mathbf{G}_{\mathrm{KJI}}^{2})_{\alpha}+(\mathbf{G}_{\mathrm{KIJ}}^{2})_{\alpha}\right]\right\}\\
&\times\left\{\prod_{\beta=\mathrm{x,y,z}}\prod_{\mathrm{I}=1}^{\mathrm{N}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}m_{\mathrm{i}}(r_{\mathrm{i},\beta}-\mathrm{R}_{\mathrm{lab},\beta}^{\mathrm{I}})\right)-\mathrm{W}_{\mathrm{I},\beta}\right]\right\}\left\{\prod_{\mathrm{I,J}=1}^{\mathrm{N}}\delta\left[\left(\sum_{\mathrm{i}=1}^{\mathrm{n}}d_{\mathrm{i}}^{\mathrm{I}}d_{\mathrm{i}}^{\mathrm{J}}m_{\mathrm{i}}\right)-\mathrm{M}_{\mathrm{IJ}}\right]\right\}.
\end{aligned} \tag{A21}$$

Recalling that $\mathrm{P}_{\boldsymbol{\alpha}}\mathbf{F}_{\mathrm{K}}=\mathcal{P}\mathbf{F}_{\mathrm{K}}/\Omega$ as well as Eqs. (27), (A11), (A14), and (A17) then gives

$$\begin{aligned}
\mathrm{P}_{\boldsymbol{\alpha}}\mathbf{F}_{\mathrm{K}}=\langle\mathbf{F}_{\mathrm{K}}\rangle=&-kT\langle\sum_{\mathrm{j}=1}^{\mathrm{n}}\frac{\partial d_{\mathrm{j}}^{\mathrm{K}}}{\partial\mathbf{r}_{\mathrm{j}}}\rangle+kTe^{\beta\mathrm{V}}\sum_{\mathrm{I}=1}^{\mathrm{N}}\frac{\partial}{\partial\mathbf{W}_{\mathrm{I}}}\cdot\left(\langle\mathbf{G}_{\mathrm{IK}}^{1}\rangle e^{-\beta\mathrm{V}}\right)+kTe^{\beta\mathrm{V}}\sum_{\mathrm{I}=1}^{\mathrm{N}}\mathrm{M}_{\mathrm{KI}}\frac{\partial}{\partial\mathbf{W}_{\mathrm{I}}}e^{-\beta\mathrm{V}}+kTe^{\beta\mathrm{V}}\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\frac{\partial}{\partial\mathrm{M}_{\mathrm{IJ}}}\left[\langle\mathbf{G}_{\mathrm{KJI}}^{2}\rangle e^{-\beta\mathrm{V}}+\langle\mathbf{G}_{\mathrm{KIJ}}^{2}\rangle e^{-\beta\mathrm{V}}\right]\\
=&-kT\langle\sum_{\mathrm{j}=1}^{\mathrm{n}}\frac{\partial d_{\mathrm{j}}^{\mathrm{K}}}{\partial\mathbf{r}_{\mathrm{j}}}\rangle+kT\sum_{\mathrm{I}=1}^{\mathrm{N}}\frac{\partial}{\partial\mathbf{W}_{\mathrm{I}}}\cdot\langle\mathbf{G}_{\mathrm{IK}}^{1}\rangle-\sum_{\mathrm{I}=1}^{\mathrm{N}}\frac{\partial\mathrm{V}}{\partial\mathbf{W}_{\mathrm{I}}}\cdot\langle\mathbf{G}_{\mathrm{IK}}^{1}\rangle-\sum_{\mathrm{I}=1}^{\mathrm{N}}\mathrm{M}_{\mathrm{KI}}\frac{\partial\mathrm{V}}{\partial\mathbf{W}_{\mathrm{I}}}+kT\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\frac{\partial}{\partial\mathrm{M}_{\mathrm{IJ}}}\left[\langle\mathbf{G}_{\mathrm{KJI}}^{2}\rangle+\langle\mathbf{G}_{\mathrm{KIJ}}^{2}\rangle\right]-\sum_{\mathrm{I,J}=1}^{\mathrm{N}}\frac{\partial\mathrm{V}}{\partial\mathrm{M}_{\mathrm{IJ}}}\left[\langle\mathbf{G}_{\mathrm{KJI}}^{2}\rangle+\langle\mathbf{G}_{\mathrm{KIJ}}^{2}\rangle\right].
\end{aligned} \tag{A22}$$

Summing the expressions for $\langle\boldsymbol{\Lambda}_{\mathrm{K}}^{2}\rangle$ and $\langle\mathbf{F}_{\mathrm{K}}\rangle$ gives the conservative terms in Eqs. (32) for $d\mathbf{P}_{\mathrm{K}}/dt$. Notice that the factors of $kT\langle\sum_{\mathrm{j}=1}^{\mathrm{n}}\frac{\partial d_{\mathrm{j}}^{\mathrm{K}}}{\partial\mathbf{r}_{\mathrm{j}}}\rangle$ cancel upon this addition.

If one considers the mapping of Eqs. (10) with the choice in Eq. (13), then all terms involving spatial derivatives of $d_{\mathrm{i}}^{\mathrm{I}}$ are zero. This eliminates many terms compared with the more general derivation given above. Repeating the derivation in this case gives $\langle\mathbf{F}_{\mathrm{K}}\rangle=kTe^{\beta\mathrm{V}}\sum_{\mathrm{j}=1}^{\mathrm{n}}\sum_{\mathrm{I}=1}^{\mathrm{N}}c_{\mathrm{j}}^{\mathrm{K}}d_{\mathrm{j}}^{\mathrm{K}}c_{\mathrm{j}}^{\mathrm{I}}d_{\mathrm{j}}^{\mathrm{I}}m_{\mathrm{j}}/\mathrm{M}_{\mathrm{J}}\partial/\partial\mathbf{R}_{\mathrm{I}}e^{-\beta\mathrm{V}}=-\widetilde{\mathrm{M}}_{\mathrm{K}}/\mathrm{M}_{\mathrm{K}}\partial\mathrm{V}/\partial\mathbf{R}_{\mathrm{K}}$, again with the use of $d_{\mathrm{j}}^{\mathrm{I}}d_{\mathrm{j}}^{\mathrm{K}}=(d_{\mathrm{j}}^{\mathrm{K}})^{2}\delta_{\mathrm{IK}}$. This then leads to the expression for the force given by Eq. (20).

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